

# QUAD 34

service data

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## introduction

The Quad 34 control unit has inputs for pickup, tape recorder, radio tuner and compact disc player. The latter can also be used for a second radio tuner or for record/replay with a two head cassette recorder. The chosen input is selected by pushbuttons and amplified to power amplifier input level. Filter, Tilt and Bass controls enable the listener to correct for certain room effects and programme balance.

## circuit description

### POWER SUPPLY

Transformer secondary volts are approximately 27V depending on the value of the AC supply volts. This is rectified and smoothed, and the resulting 30V DC is applied to the input of a negative voltage regulator. The output from the regulator is  $-18V$  referenced to the positive rail for input voltages of  $-21V$  to  $-33V$ .

IC23 is connected as an earthed feedback buffer and is used in conjunction with R123 and R124 to convert the regulator output to a +ve and -ve rail. The values of the rails are determined by the values of R123 and R124. Any fluctuation in the voltage at pin 3 of IC23 will cause the Op Amp to pass current through pin 3 and the appropriate HT rail to offset the original change. C58 and C84 decouple the HT rails. R84, D30 and R85 define the  $+7.5V$  and  $-7.5V$  supply rails necessary to drive the CMOS circuits, with C57 and C59 decoupling these points.

### CLAMP CIRCUIT

Immediately after switch on of the 34 the regulator will be in the voltage drop-out mode since there are insufficient input volts. Under these conditions there will be a 1.1V drop between the regulator output and input. This is applied via the differentiator of R102 and C69 to the base of T13 which will remain firmly off. Point A (see circuit diagram) will be +ve with respect to earth, hence the clamp transistors T14 and T15 will be turned on short circuiting the audio output to earth.

When the voltage applied to the input of the regulator is sufficient to operate the regulator further increase in input will give no change in output increasing the voltage applied to the differentiator. C69 will continue to charge via R102, allowing time for the HT rails to stabilise before switching on T13. When T13 turns on, the voltage at point A will drop and the clamp transistors will be turned off. The value of reversed bias applied to the base emitter junctions of the clamp transistors will vary according to the AC supply voltage however D32 ensures that the value never exceeds approximately 4V.

**Note:** The difference between regulator input and output voltage must be approximately equal to 2V before T13 will turn on hence turning off the clamp transistors.

Upon switch off of the 34, C74 will discharge slowly enabling the regulator to sustain o/p voltage. As C74 discharges the regulator input voltage will fall (become more positive) and C69 will discharge through D31. When the regulator input voltage comes within 2V of the output voltage (i.e. input approx.  $-20V$ ) T13 will turn off, turning on the audio clamp transistors.

### SWITCHING

Fig 1 is a schematic diagram illustrating how the electronic switching controls the signal flow through the 34. Switching is carried out using logic circuits to control analogue switches.

Disc, Radio and CD (Aux) inputs are connected to a common signal bus via analogue switches. Tape record is permanently connected to the signal bus. Selection of Disc, Radio or CD (Aux) will enable signal from the appropriate input onto the signal bus, hence Tape record. The other two inputs will be cancelled. Signal from the bus is only transmitted onto the main signal path through the 34 if Tape replay is not selected. Selection of Tape replay overrides the signal bus enabling Tape input onto the main signal path, thus providing the monitoring facility.

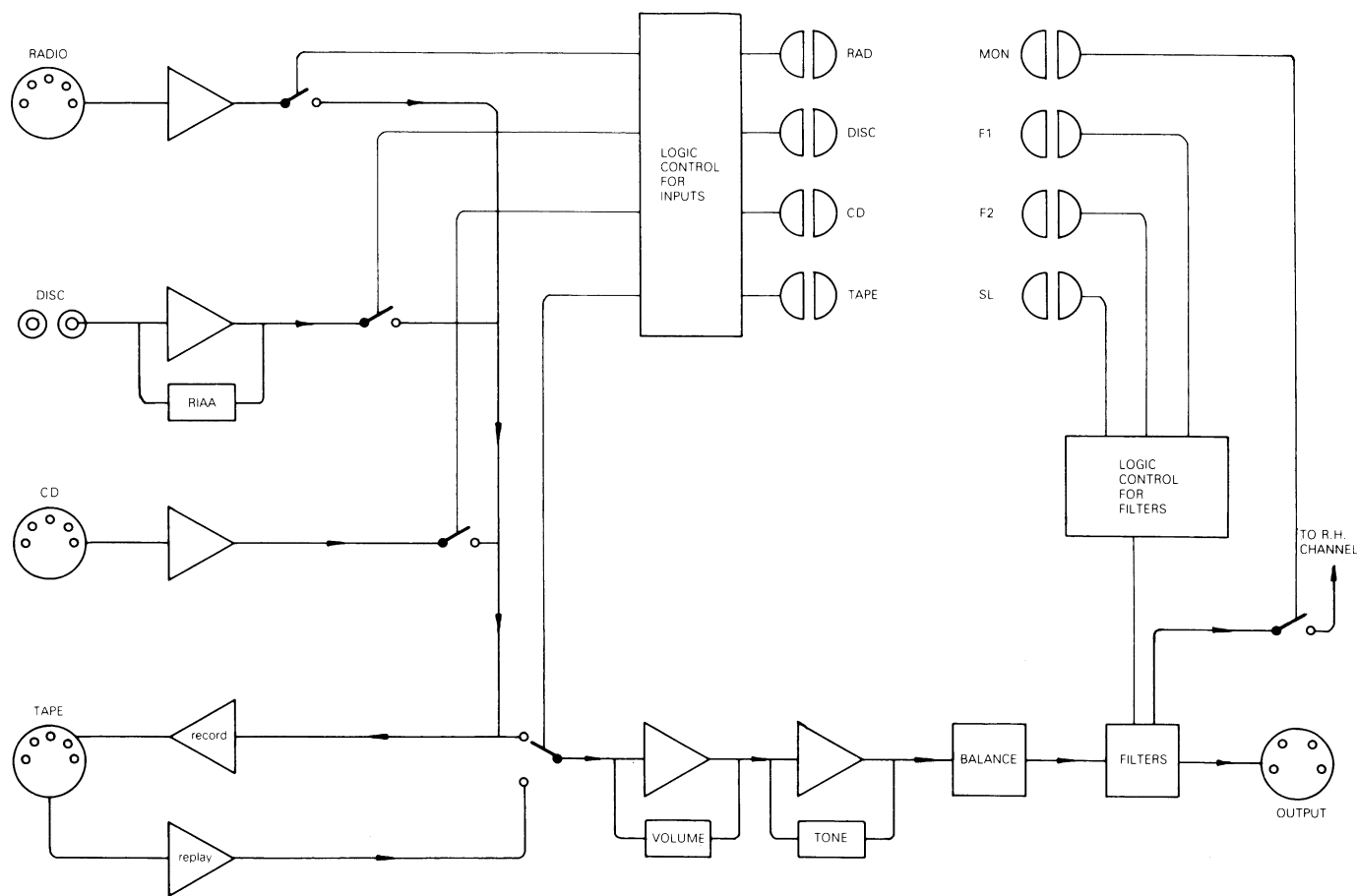


Fig. 1 SCHEMATIC SIGNAL FLOW DIAGRAM

Disc, Radio and CD

Control logic of the Disc, Radio and CD (Aux) analogue switches is based around the triple NOR gate latch of IC11.

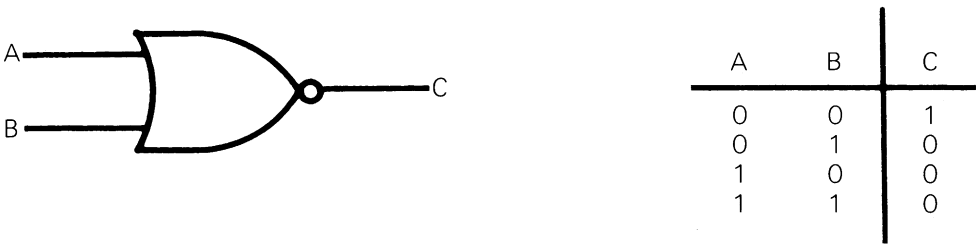


Fig. 2

The output of the NOR gates are used to control the analogue switches of the Disc, Radio and CD (Aux) inputs. The inputs to each NOR gate are the outputs from the other two NOR gates. Selection of, say, Radio will result in an instantaneous logic 1 being applied to pin 10, thence to the control pins of the Radio analogue switches. Also the logic 1 will be applied to the inputs of the Disc and CD NOR gates, (pins 13 and 5) making their respective outputs zero. The two zero's are applied to the Radio NOR gate inputs, pins 8 and 9, hence pin 10 the Radio control pin will sustain logic 1. The logic 1 is also used to turn on indicator led D16 via T6. The process is similar for selection of CD and Disc. The differentiator action of C44 and R62 ensures that Radio is automatically selected upon switch on of the 34.

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## Tape

Operation of the tape switch is based around a D-type flip flop (IC14). This has complementary outputs Q (pin 13) and  $\bar{Q}$  (pin 12). Data on pin 9 is transmitted to pin 13 on each positive clock (pin 11) transition. Connected as shown, data is determined by the o/p on pin 12,  $\bar{Q}$ . Assuming initial conditions of Q (pin 13) at logic high then  $\bar{Q}$  (pin 12) will be at logic low. The outputs will enable two of the analogue switches of IC1 (pins 3, 4, 5 and 6, 8, 9) and disable the other two (pins 1, 2, 13 and 10, 11, 12), thus selecting tape input. Also D18 will illuminate via T8.

Further selection of Tape will apply a positive transition to clock, (pin 11), and data on pin 9 set by  $\bar{Q}$  (pin 12) will be transmitted to pin 13, hence the outputs change their states. Under these conditions the analogue switches of IC1 will be enabled in the opposite way to the previous configuration. Signal from the common bus (Disc, Radio or CD) is then transmitted through the set and Tape replay is disabled. Selection of Tape again will cause the outputs to change states and once again Tape replay is enabled.

Initial conditions are defined by the differentiator action of R83 and C55. Upon switch on a short positive pulse is applied to the reset pin (pin 10) of the flip flop. This sets  $\bar{Q}$  (pin 12) to logic high hence ensuring tape is not selected initially.

## Mono

Selection of mono links two points in the left and right channel circuits by means of an analogue switch. The switch is controlled in the same way as for selection of tape using a D type flip flop (see above).

## Filter Switching

Filtering is controlled by analogue switches within the filter circuit. Again the control logic is based around D type flip flops which are used in much the same way as described for Tape selection (see above). Selection of Filter 1 will enable the appropriate analogue switches in the filter circuit via the D type flip flop. Also the select pulse will be applied to the reset pin of the D type flip flop associated with Filter 2, hence ensuring that selection of Filter 1 cancels Filter 2. Selection of Filter 2 will cancel Filter 1 in an identical manner. Initial conditions are defined by the differentiator action of C68, R100 and C79, R108. Upon switch on the action of these components is to apply a positive pulse to the reset pins of the two flip flops, ensuring that neither Filter 1 or Filter 2 is selected.

Slope operation is again based around a D type flip flop controlling an analogue switch within the filter network. However since Slope can only be selected if Filter 1 or Filter 2 is selected and Slope is automatically selected with no led indication (see filters) when neither Filter 1 or Filter 2 is selected then additional circuitry is employed to detect and enforce these conditions.

When Filter 1 and Filter 2 are off, the  $\bar{Q}$  outputs of their respective flip flops (pin 12 of IC15 and pin 2 of IC16) are at logic high. Neither D23 or D24 will conduct so the reset pin of the Slope flip flop (pin 10) will be at logic high. This leaves the flip flop locked in the off mode regardless of whether the Slope button is selected or not. However the logic high on pin 10 is applied to the control pin of the steep slope analogue switch via D29. Under these conditions Slope is automatically selected but the Slope indicator led remains off. This provides a rapid tail off in frequency response above the audio band.

When either filter is selected then the  $\bar{Q}$  output of the associated flip flop will be at logic low and either D23 or D24 will conduct. This will set the reset pin of the Slope flip flop (pin 10) at logic low enabling Slope to be selected. Operation of the Slope switching circuit is then the same as the Tape input switching circuit described above. Initial conditions are defined by the states of Filters 1 and 2, which are set to be off upon switch on of the 34.

## TONE CONTROLS

The Tilt and Bass circuits of the 34 are based around one Op Amp, per channel, IC13 in the case of the right hand channel, and IC12 for the left hand channel.

## Tilt Circuit

The function of the Tilt circuit is to provide a gradual change in output level across the frequency spectrum, centered on 0dB, (see fig 3).

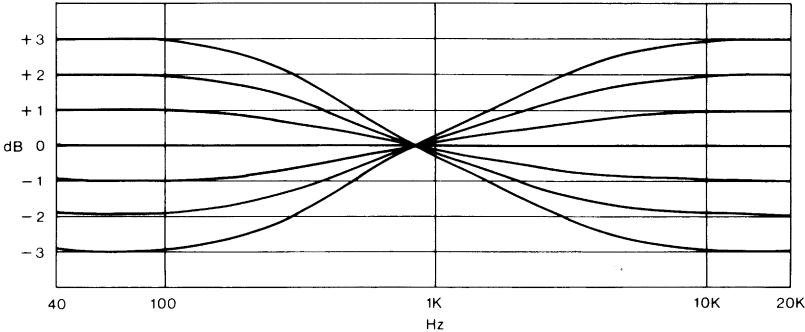


Fig. 3

The Tilt circuit may be built up in three stages.

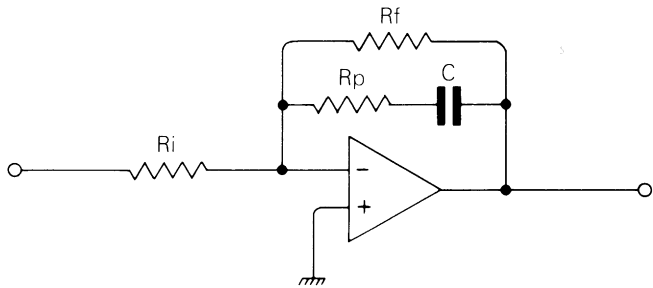
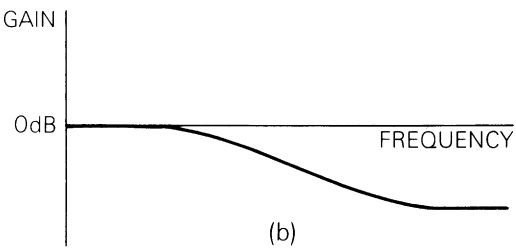


Fig. 4

(a)



(b)

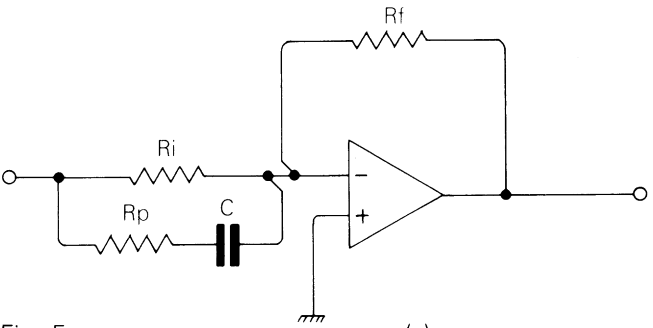
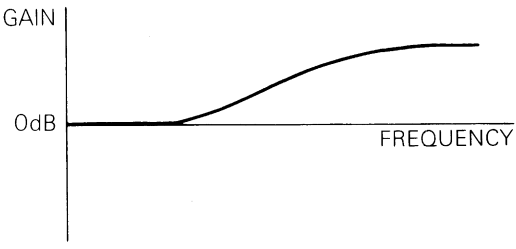


Fig. 5

(a)



(b)

1. In the circuit shown in fig 4(a) gain is determined by;
- At low frequency  $\frac{R_f}{R_{in}}$
- At high frequency  $\frac{R_f//R_p}{R_{in}}$

If  $R_f = R_i$ , this would give a frequency response as shown in fig 4(b). Similarly the frequency response for the circuit shown in fig 5(a) is as shown in fig 5(b).

2. If the two circuits of figs 4 and 5 were added together the resulting frequency response would be flat at 0dB. If a pot is then added as shown in fig 6 (a) the response would only remain flat if the pot were in the central position. Displacement of the pot to either of the end extremes would yield one of the two responses shown in fig 6(b).

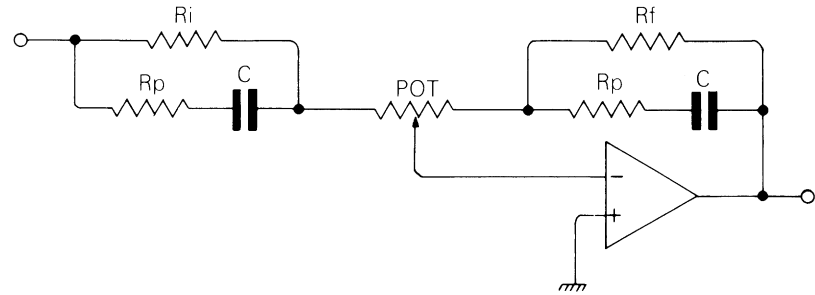
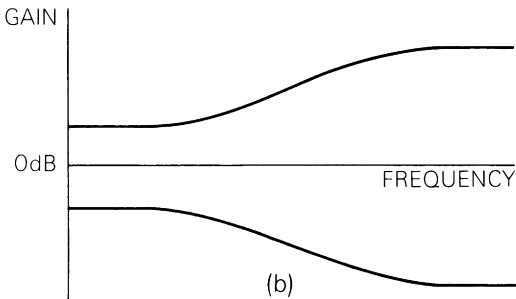


Fig. 6

(a)



(b)

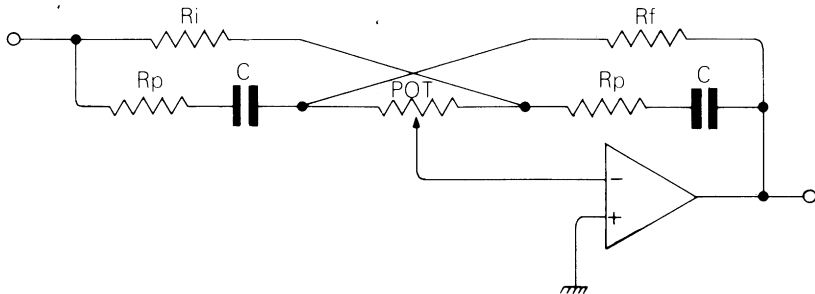
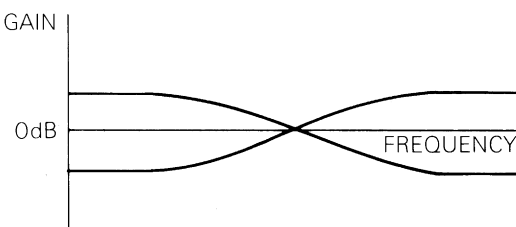


Fig. 7

(a)



(b)

3. Crossing of  $R_f$  and  $R_{in}$  as shown in fig 7(a) results in the two most extreme responses being as shown in fig 7(b).

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## Bass Circuit

The Bass control uses two separate circuits for lift and step modes. For this reason the resistance track in the Bass pot is discontinuous as shown in fig 9(a).

## Bass Lift

Fig 8(a) shows a simplified bass lift circuit. The Tilt network here is represented by ZT1 and ZT2 which are equal when the Tilt switch is in the zero position. At medium frequencies the capacitive reactances are negligible and the gain is set by ZT1 and ZT2. At low frequencies the reactance of Co becomes significant and, if R bass is selected, reduces the negative feedback hence increasing the gain. R bass simply varies the frequency at which the turnover point occurs. Cin provides a second turnover point further down the spectrum thus defining the frequency at which the gain is maximum and ultimately reducing the gain to unity. The resulting curves are shown in fig 8(b).

## Bass Step

Bass step is provided by the low frequency potential divider of C37, C39, R48 and RV3(b), (in the case of the right hand channel). Variation of RV3(b) will set the frequency at which attenuation begins. C37 and C39 set the ultimate attenuation to approx. -6dB. This value is slightly modified by R48.

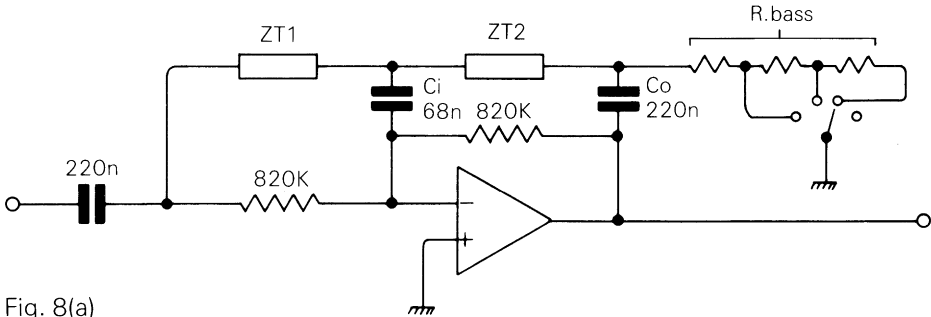


Fig. 8(a)

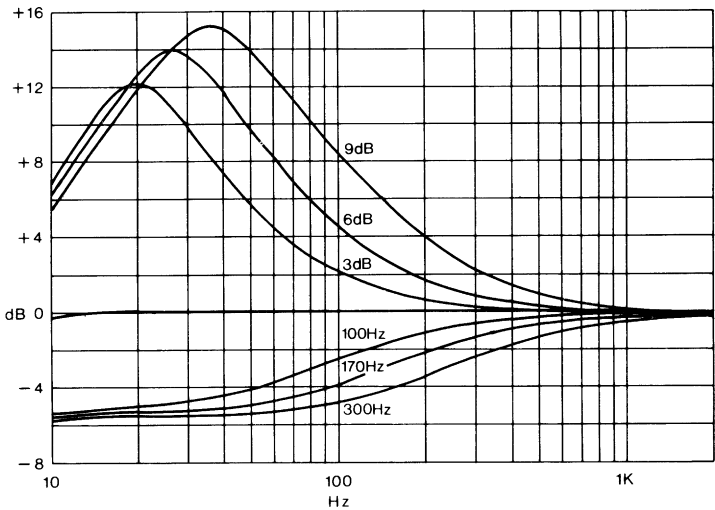


Fig. 8(b)

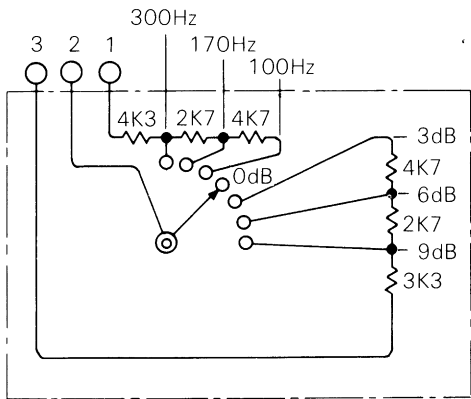
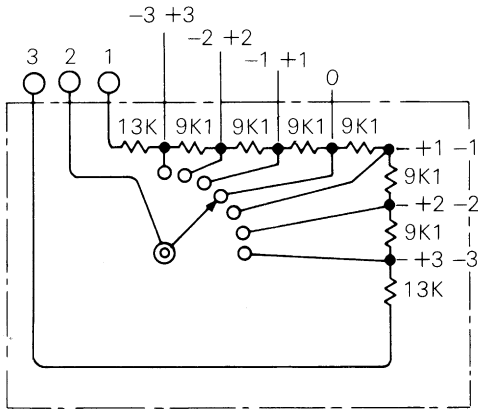
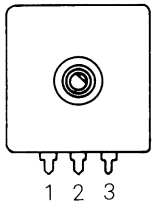


Fig. 9 (a) BASS POT



(b) TILT POT

FILTERS

The filter circuit is designed to cater for 5 different variables, these being no filters, Filter 1 only, Filter 2 only, Filter 1 Slope and Filter 2 Slope.

**Note:** When neither filter is selected Slope is automatically selected but this is not indicated by the slope led, (see Switching page 4). This rapid tail off in frequency response discriminates against frequencies above the audio band. When Filter 1 or Filter 2 is selected the upper cut off frequency is lowered (see fig 10). The rate of attenuation will be 6dB/octave.

Selection of Slope with either filter increases the slope to 12dB/octave. Also with Slope selected the upper cut of frequency (−3dB point) for a given filter is modified so that filtering attenuation begins at the same point.

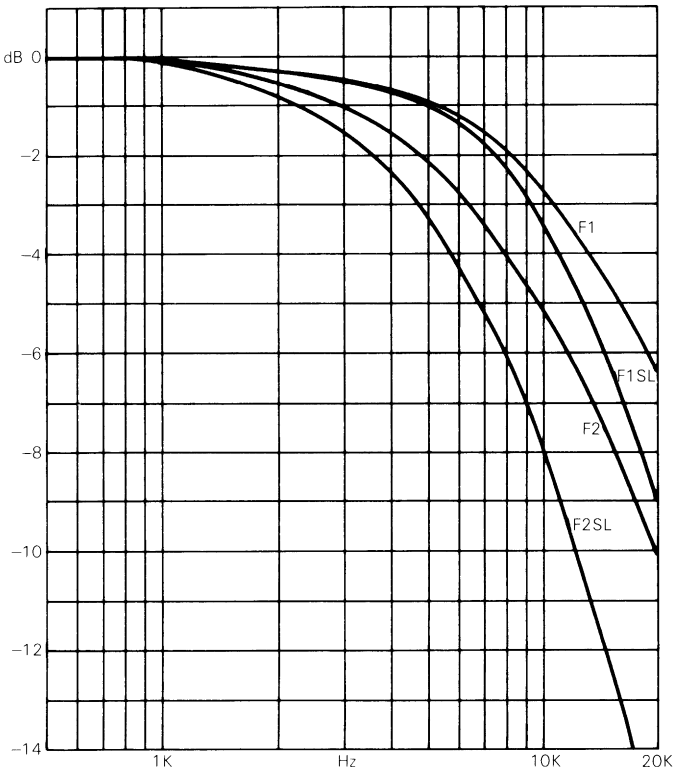


Fig. 10

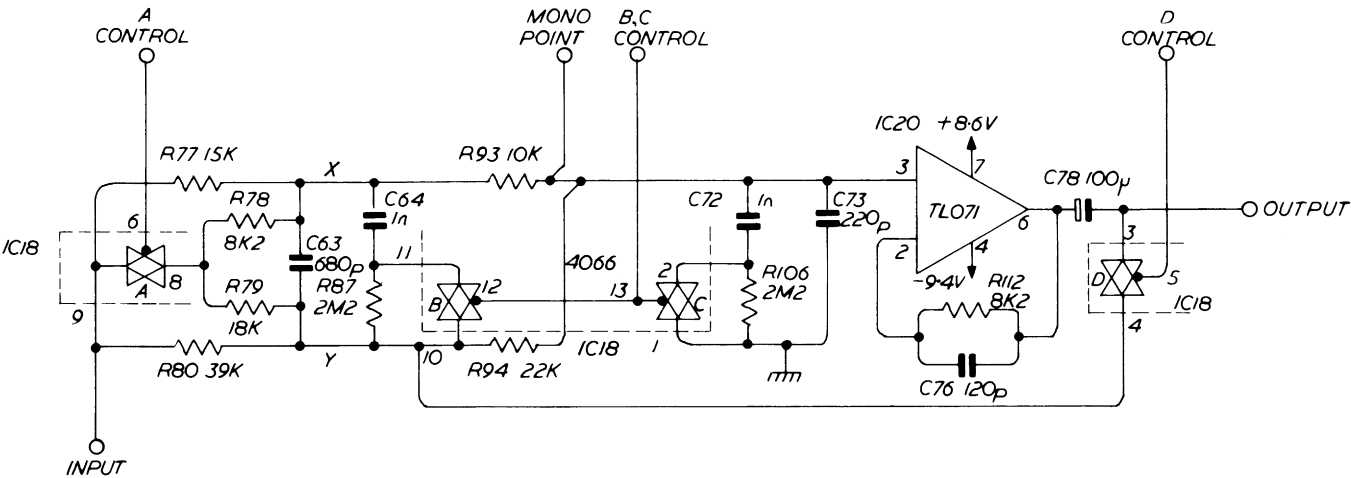


Fig. 11

These variations are achieved using logic to control analogue switches within the filter circuit, (see fig 11). For 6dB octave filtering the circuit is connected as a buffered R-C filter (12b). For 12dB/octave filtering (Slope) the circuit is connected in a Sallen-Key configuration as shown in fig 12(a).

The values of R and C in either network are determined using the four analogue switches to connect (in the case of the right channel shown in fig 11) R77, R78, R79, R80, R93, R94, C63, C64, C72 and C73 in various configurations. (Similarly for the left hand channel).

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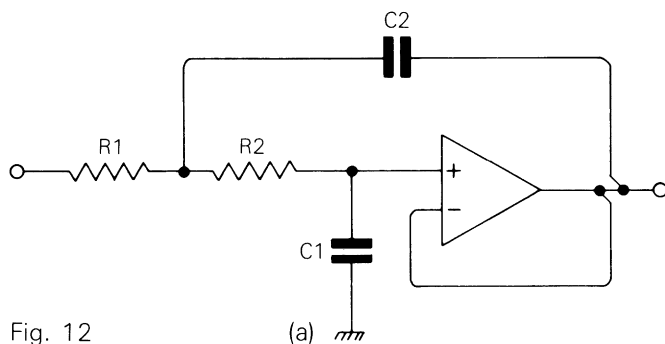
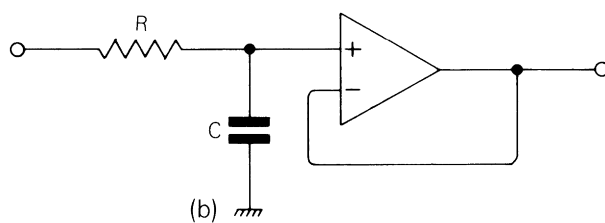


Fig. 12



## 1. Filter 1

With Filter 1 selected and no slope, the analogue switch 'D' (fig 11) is disabled and A B and C are enabled.

- (i) Analogue switch A connects R77 in parallel with R78 and R79 in parallel with R80. These resistors together with R93 and R94 form a balanced bridge circuit with no potential difference across C63, hence this component plays no active part.
- (ii) Analogue switch B connects C64 across C63 hence this component also plays no active part.
- (iii) Analogue switch C connects C72 in parallel with C73. The equivalent circuit is shown in fig 12(b) where;

$$R = ((R77//R78) + R93)/((R79//R80) + R94)$$

$$C = C72 + C73$$

## 2. Filter 2

Conditions are much the same as in 1 except that analogue switch A is no longer enabled therefore R78 and R79 are left out of circuit.

The equivalent circuit is shown in fig 12(b) where;

$$R = (R77 + R93)/(R80 + R94)$$

$$C = C72 + C73$$

## 3. No Filters

Analogue switches A and D only are enabled.

- (i) Analogue switch A makes connections as described in 1(i).
- (ii) Analogue switch D switches in the feedback circuit. This makes the effects of R94, R79 and R80 negligible. The circuit becomes a sallen key filter, the circuit for which is shown in fig 12(a) where;

$$R1 = R77//R78$$

$$R2 = R93$$

$$C1 = C73$$

$$C2 = C63$$

## 4. Filter 1 with slope

All analogue switches are enabled.

- (i) Analogue switches A, B, C make connections as described in 1(i), (ii) and (iii) respectively.
- (ii) Analogue switch D makes connections as described in 3(ii). The circuit becomes a sallen key filter as shown in fig 12(a) where;

$$R1 = R77//R78$$

$$R2 = R93$$

$$C1 = C72 + C73$$

$$C2 = C63 + C64$$

## 5. Filter 2 with slope

The conditions are much the same as in 4 except that analogue switch A is no longer enabled. The equivalent circuit is as shown in fig 12(a) where;

$$R1 = R77$$

$$R2 = R93$$

$$C1 = C72 + C73$$

$$C2 = C63 + C64$$

# fault finding

Fault finding is best carried out using the flow diagram shown in fig 13. It should be noted however that the diagram caters for single fault conditions only and is not suitable for diagnosing multiple faults.

Exit from the diagram will be at one of the 35 diagnoses, each of which is clearly defined below. Since the diagram is based on likely or recurring faults some of the diagnoses are arrived at by default. This form of reasoning may occasionally provide an incorrect answer. Always check that the diagnosis seems reasonable by looking at the circuit diagram.

Reference is made in the diagram to isolating the regulator o/p and measuring the o/p voltage. This is best done by removing the regulator and resoldering it to the rear of the pcb, leaving the o/p pin disconnected. The o/p voltage can then be measured between the o/p pin and the +ve 8.6V HT rail, and should be -18V with respect to this rail. Where it is necessary to measure HT voltages, see pages 15 and 16.

## DIAGNOSES

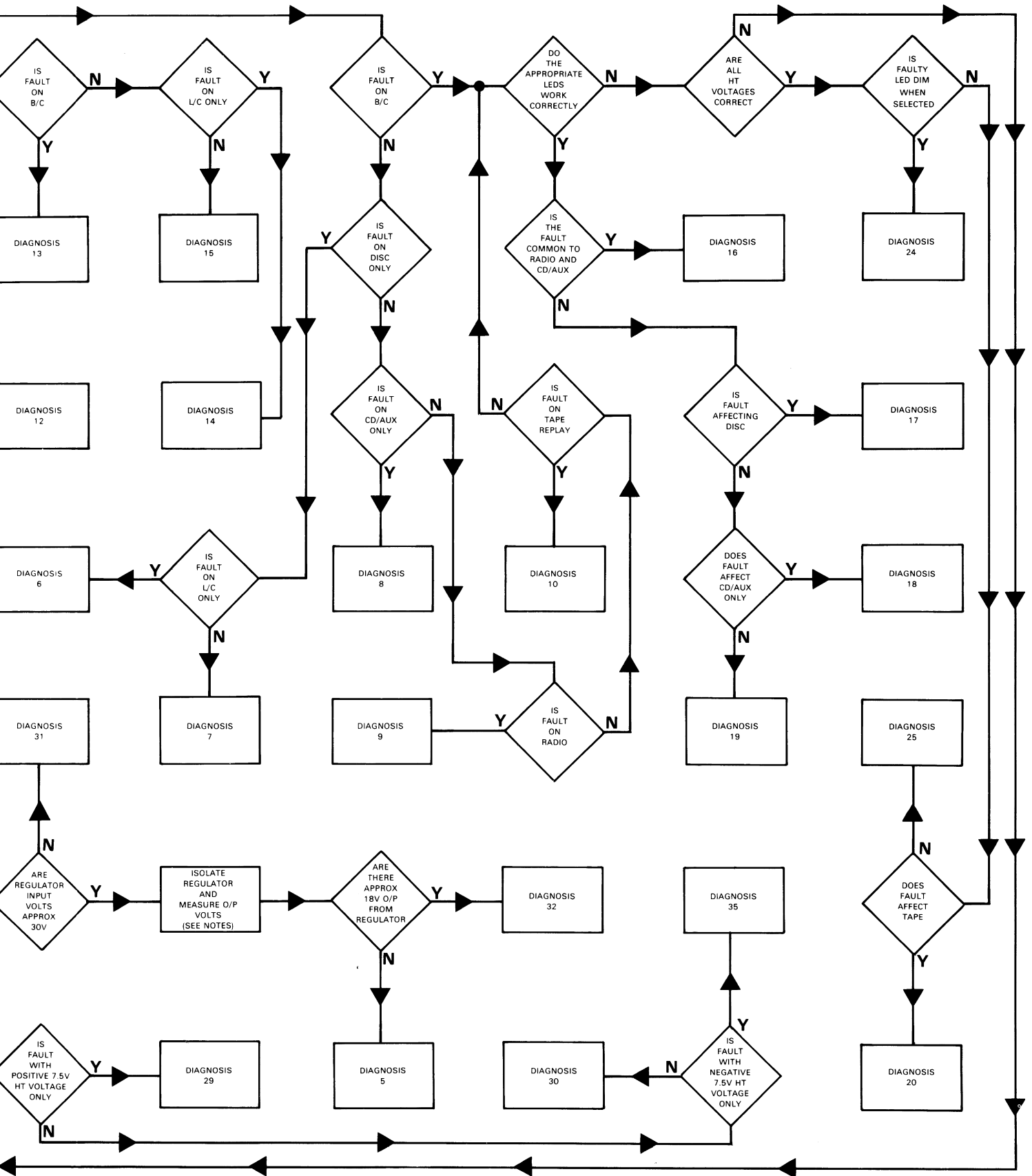
1. IC18.
2. Fuse blown due to spurious AC power supply spike.
3. (i) Check that the voltage selector is set correctly.  
(ii) L1 (Transformer).
4. This fault is most likely to occur with pre-serial number 2000 34's. Replace with a new transformer which will have two electrically redundant tags. These should be cut off after which the transformer may be located on the p.c.b. The remaining tags should be uncut and bent under the p.c.b. to increase mechanical rigidity.
5. IC21 U/S.
6. (i) IC7 U/S.  
(ii) T1, T2 U/S.  
(iii) IC22 U/S.  
(iv) Ribbon Cable O/C.
7. (i) IC8 U/S.  
(ii) T3, T4, U/S.  
(iii) IC22 U/S.  
(iv) Ribbon Cable O/C.
8. (i) For 34's post S/N 8000 IC24 U/S.  
(ii) IC2 U/S.
9. IC2 U/S.
10. (i) For 34's post S/N 8000 IC26 U/S.  
(ii) IC1 U/S.
11. (i) For 34's pre S/N 8000 IC3 U/S.  
For 34's post S/N 8000 IC25 U/S.
12. (i) For 34's pre S/N 8000 IC4 U/S.  
For 34's post S/N 8000 IC25 U/S.
13. (i) D32 U/S. Particularly on 34's pre S/N 6000.  
(ii) IC1 U/S.  
(iii) For 34's post S/N 8000 IC27 U/S.
14. (i) For 34's pre S/N 8000 IC5 U/S.  
For 34's post S/N 8000 IC27 U/S.  
(ii) IC19 U/S.  
(iii) IC12 U/S.  
(iv) IC9 U/S.  
(v) D/J volume pot.
15. (i) For 34's pre S/N 8000 IC6 U/S.  
For 34's post S/N 8000 IC27 U/S.  
(ii) IC20 U/S.  
(iii) IC13 U/S.  
(iv) IC10 U/S.  
(v) D/J volume pot.

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16. IC2 U/S.
17. (i) Check that disc board edge connector is properly located.  
(ii) IC22 U/S.
18. For 34's post S/N 8000 IC24 U/S.
19. For 34's post S/N 8000 IC26 U/S.
20. IC14 U/S.
21. For 34's post S/N 8000 IC25 U/S.
22. A touch hot IC is an indication that the IC itself is faulty.
23. Associated BC183 U/S.
24. Associated BC183 U/S.
25. IC11 U/S.
26. (i) Associated BC183 U/S.  
(ii) Led U/S.
27. IC15 or IC16 U/S.
28. IC17 U/S.
29. CMOS IC S/C. To isolate the faulty IC unsolder pin 14 from IC1, IC2, IC11, IC14, IC15, IC16, IC17, IC18 and IC22 in turn, using a process of elimination to determine the faulty component.
30. Diagnosis is as for 29 except that both pins 14 and 7 should be disconnected.
31. (i) L1 (transformer).  
(ii) D34.
32. High current Op Amp. Isolate HT pins of each Op Amp in turn using a process of elimination to isolate the faulty component.
33. IC14 U/S.
34. IC22 U/S.
35. Diagnosis is as for 29 except that pin 7 of the IC's listed should be disconnected.



Fig. 13



# test procedure

Complete testing of a 34 may most easily be carried out by using the equipment listed below and interconnecting these units as illustrated in fig 14. Any signal cables used for interconnecting should be screened. The 34 cover should be removed for initial tests. Reference is made to testing on 240V, clearly the same results should be expected if testing on 110V though the current consumption will be larger.

## TEST EQUIPMENT

- Sine/Square A.F. Signal Generator
- AC Microvoltmeter
- Oscilloscope
- Anti RIAA Network (as shown in fig 18)
- Mono Amplifier
- Headphones or Loudspeaker
- 4 x 5 pin Din Plugs
- 2 x Phono Plugs
- Voltmeter

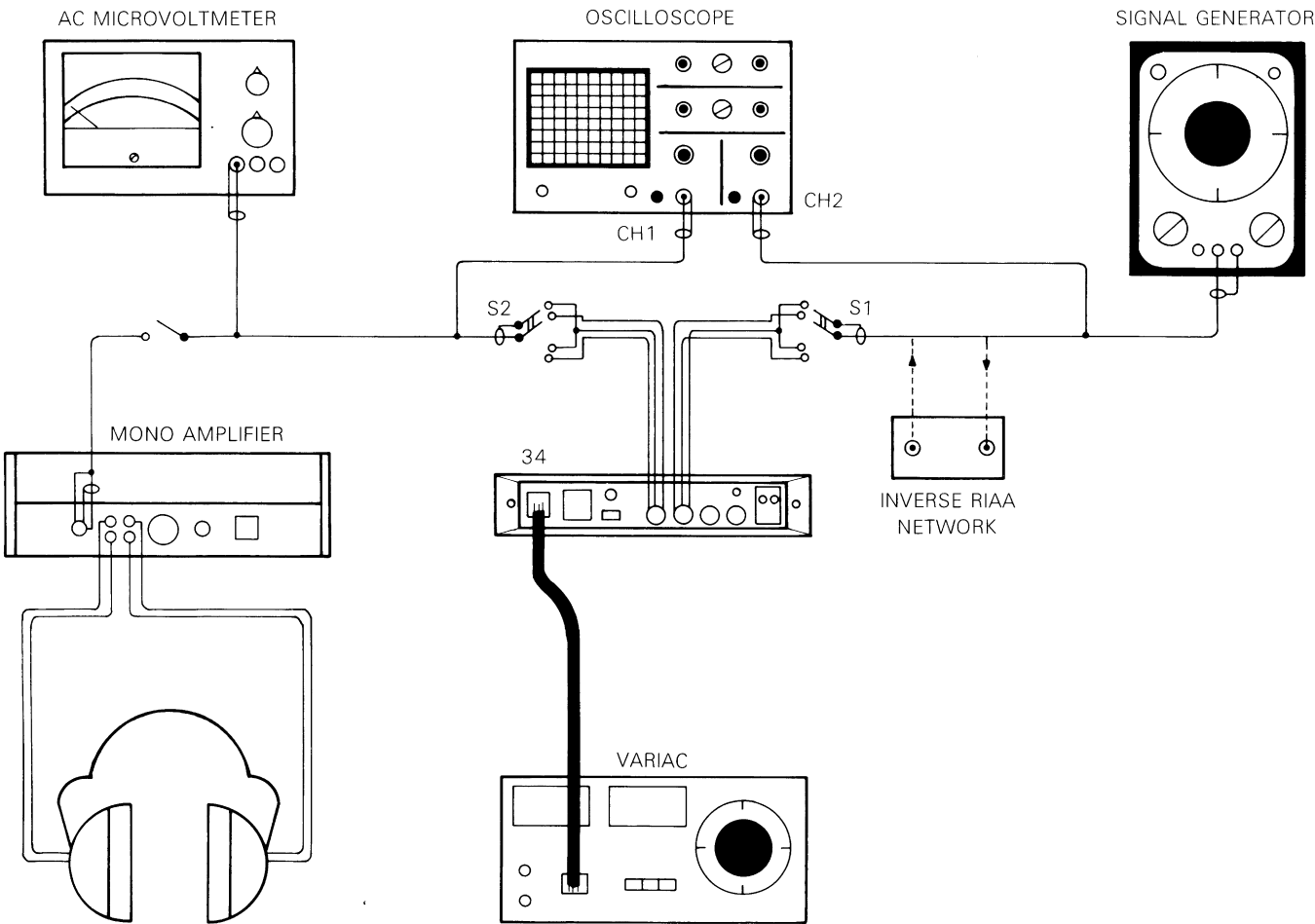


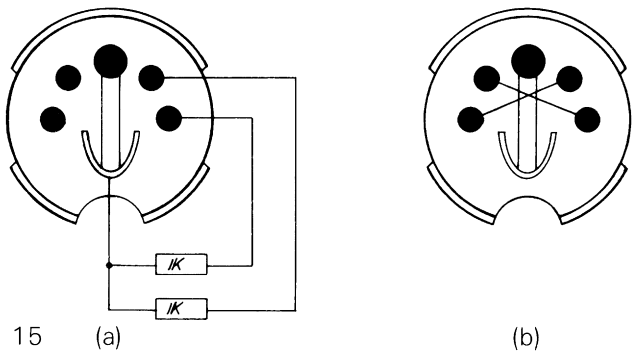
Fig. 14

The test equipment should be earthed from one localised common earth point to minimise the effects of earth loop hum.

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## Din Plugs

Three of the Din plugs should be made up to simulate 100mV loads, to be used in conjunction with noise testing. This is done by connecting two resistors as shown in fig 15(a). The other Din plug should be used to link the record pins of the tape socket to the replay pins such that the right channel record is connected to the left channel replay and vice versa as shown in fig 15(b). Short circuit each of the phono plugs.



## Connections

S1 to Radio left channel input  
S2 to left channel output

## Controls

- Variac: output volts – zero
- 34: Power – on  
Bass, Tilt – zero  
Balance – central  
Volume – 21  
Tape replay sensitivity – 300mV  
Tape record level – 300mV  
CD sensitivity – 500mV
- Oscilloscope: Ch1 – 0.5V/cm  
Ch2 – 0.1V/cm  
Timebase – 0.5ms/cm
- Signal Generator: Mode – Square Wave  
Frequency – 650Hz  
Peak to Peak Voltage – 0.2V
- Microvoltmeter: Range – suitable for 500mV reading  
Bandwidth – 10Hz – 100kHz

## 1. POWER

Increase the output of the variac to 240V whilst observing the current consumption of the 34 which should not exceed 15mA (typically 10mA). The power outlet socket should be checked by connecting to it a suitable piece of equipment. Power to the equipment should then be controlled by the 34 power switch. Remove the plastic power supply cover. Measure the 4 supply rails which should be +8.6, +7.5, –7.5 and –9.4V as shown on the diagram below. Replace the plastic power supply cover.

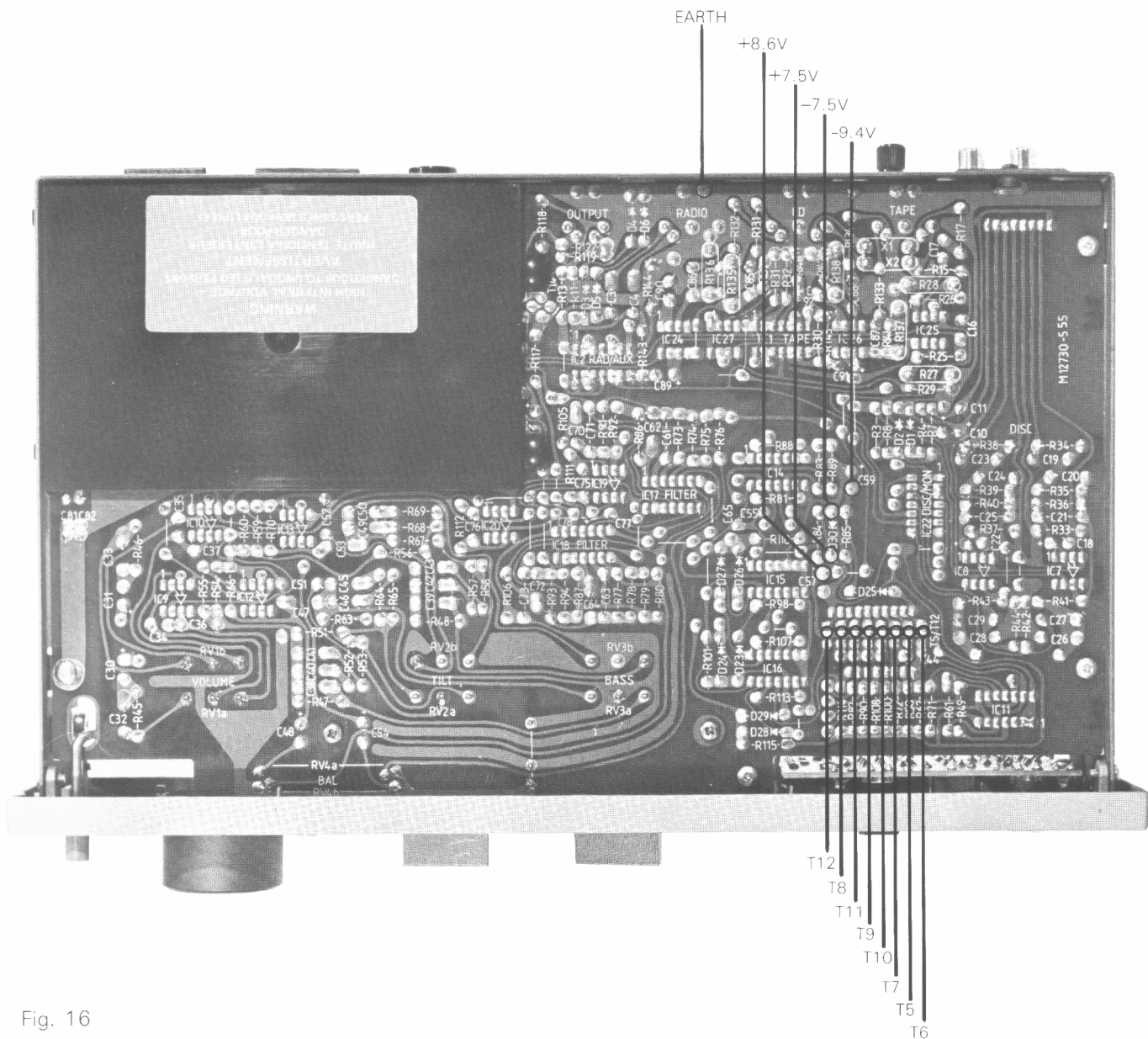


Fig. 16

2. OUTPUT

Observe the oscilloscope Ch1 beam which should show a square waveform trace as shown in fig 17. Switches S1 and S2 should then be connected so as to drive the right channel input and monitor the right channel output, when a similar waveform should be displayed. Some square waves may show some overshoot. This is due to the >20kHz Slope filter (see page 6).

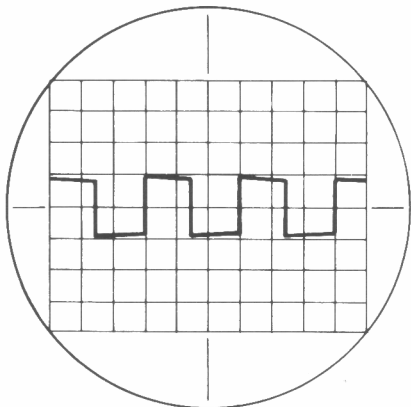


Fig 17

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### 3. CHANNEL BALANCE

Move the balance slider towards the bottom of its travel, the output waveform should progressively increase to a maximum at the bottom extreme. Move the balance slider upwards and signal should progressively decrease to zero at the top extreme.

Drive the left channel input and monitor the left channel output where the opposite to this should be obtained. Select mono and move the balance slider to the upper extreme. The waveform peak to peak value should be at a maximum of 0.6V decreasing to zero as the balance slider is moved to the lower extreme. Drive and monitor the right channel where the opposite to this should be obtained.

Select stereo mode and 10kHz sine waves at 100mV RMS. With the balance slider in the central position note the output level reading on the AC microvoltmeter. Move the balance slider to the lower extreme, the output level should increase by approx 2.5dB. Move the balance slider to the upper extreme, the output level should fall by approx 65dB.

Drive and monitor the left channel where the opposite to this should be obtained.

Centralise the balance slider.

### 4. INPUT SELECTOR AND OUTPUT LEVEL

In the following tests 'full output' refers to the waveform shown in fig 17. For 'no output' waveforms may be spikey due to the finite level of crosstalk. Select 650Hz square waves with peak to peak value at 0.2V.

### RADIO AND TAPE

- (a) Drive and monitor the left channel to obtain a full output waveform. Monitor the right channel to obtain no output. Drive and monitor the right channel to obtain full output. Monitor the left channel to obtain no output.
  - (b) Insert the Tape link plug into the Tape input socket. Select Tape and drive and monitor the right channel to obtain no output. Monitor the left channel to obtain full output. Drive and monitor the left channel to obtain no output. Monitor the right channel to obtain full output.
- Failure to achieve the correct output level suggests that Tape record and replay levels are not set to the same value. Press Tape, to re-select Radio.

### CD/Auxiliary

Remove the signal input lead from the radio input and plug this into the CD/Auxiliary socket and select this input. For units which are pre-serial number 8000 with Auxiliary input sensitivity of 100mV repeat test (a) as carried out for Radio. For post-serial number 8000 units with Auxiliary/CD input sensitivity of 500mV increase the signal generator output to 1.8V peak to peak and repeat test (a) as carried out for Radio.

### Disc

Remove the signal input lead from the Auxiliary input. The output from the signal generator should be connected to S2 via the anti RIAA circuit. In order to overcome the 75dB attenuation of the anti RIAA circuit, the input to this circuit should be set as shown below.

| Disc module | Signal generator o/p level |
|-------------|----------------------------|
| 100μV       | 4V peak to peak            |
| 200μV       | 8V peak to peak            |
| 3mV         | 22V peak to peak           |

It may be necessary to amplify the signal generator output to achieve these levels. For M/c disc input drive from the low Z output of the anti RIAA network. Fitting of 34 cover will reduce hum.

Select Disc and repeat test (a) as carried out for Radio.

### INVERSE RIAA NETWORK

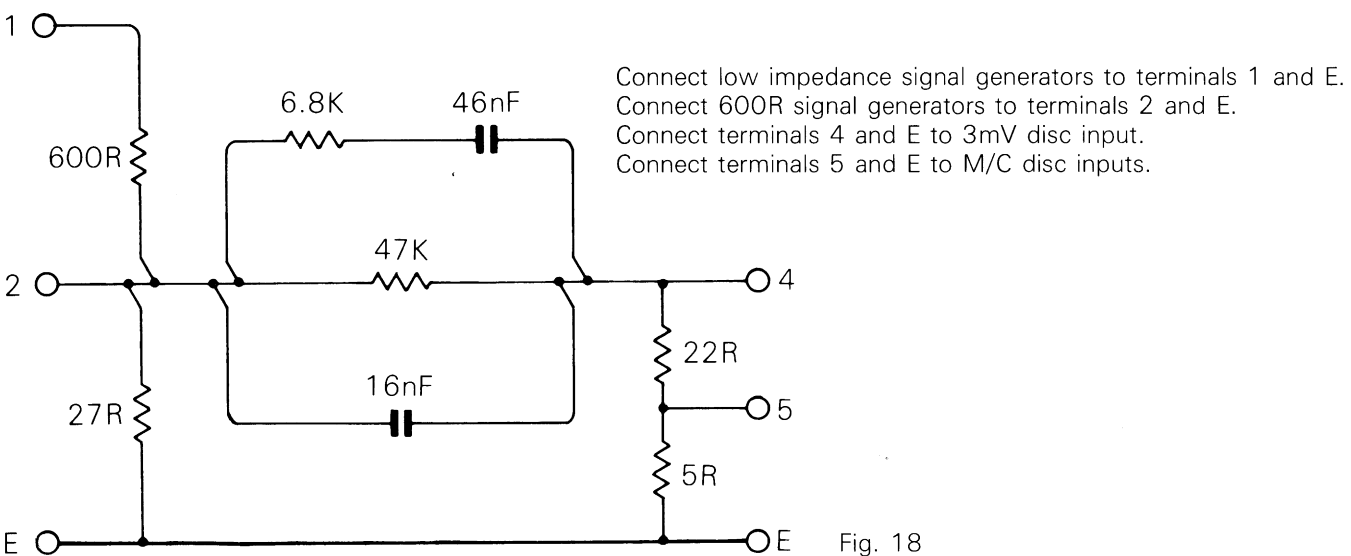


Fig. 18

Disconnect the RIAA circuit.

5. TONE AND FILTERS

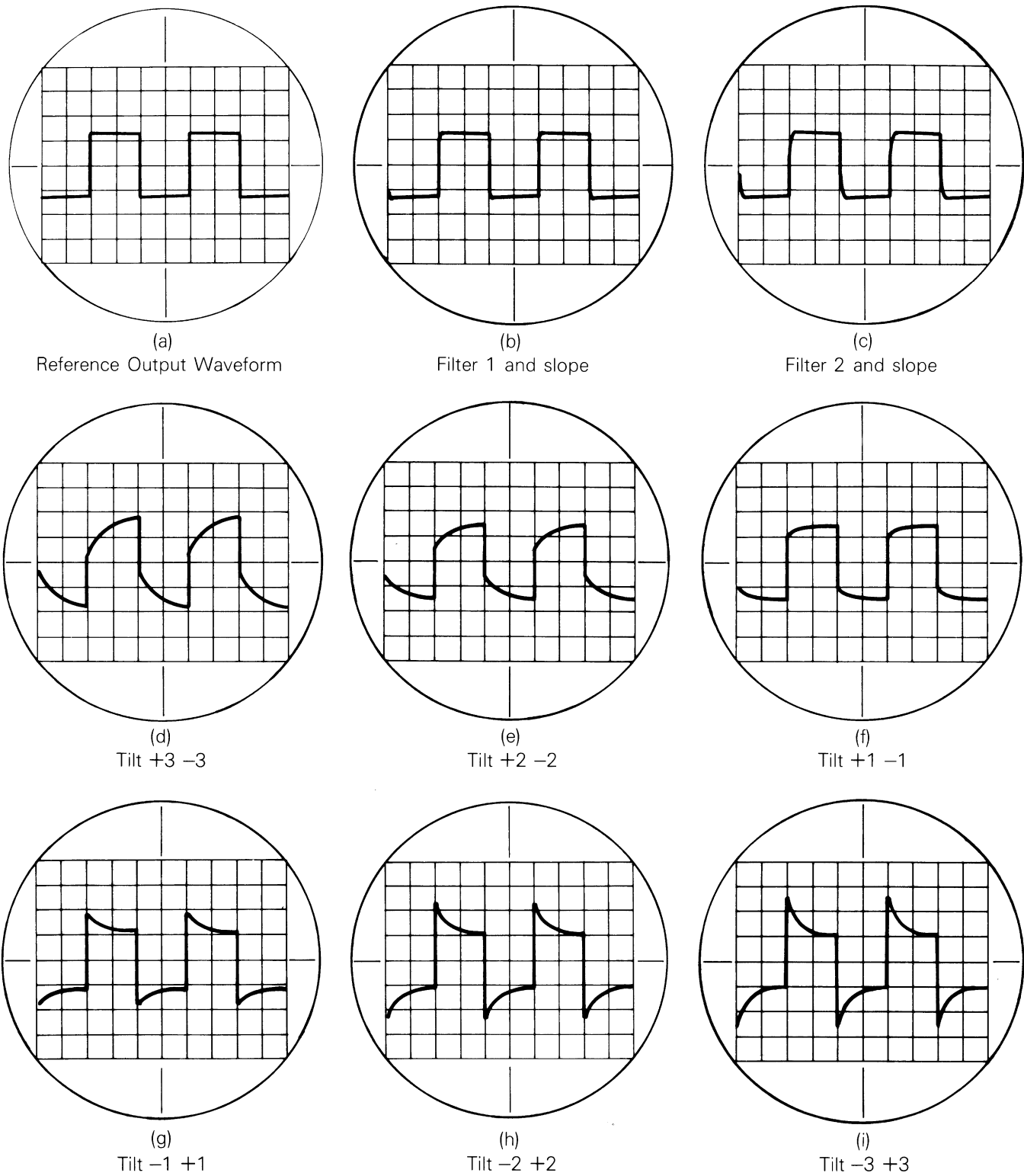
Remove the signal input lead from the Disc input. Adjust the signal generator output level to 0.2V peak to peak and apply this direct to S2. Plug the signal input lead into the Radio input and select Radio, and vol. 18.

Fig 19 shows how the output waveform is modified for each of the tone and filter settings.

Fig 19 was drawn with the timebase on the 2mV/cm range and sensitivity on the 0.2V/cm range but these were in the uncalibrated mode to illustrate more clearly the tone and filter effects. Fig 19 (a) shows the reference output waveform.

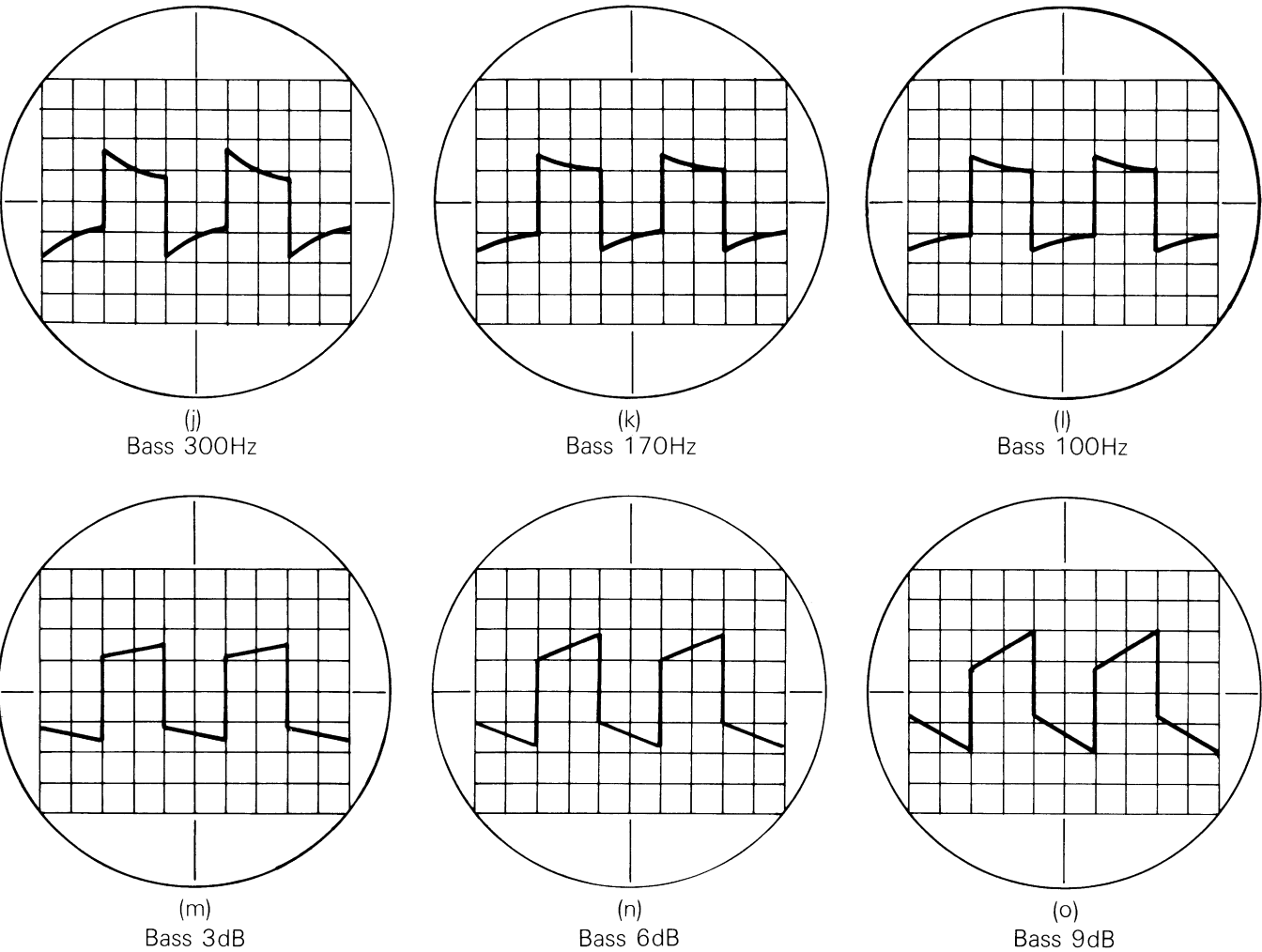
The checks should be carried out on both channels.

Fig. 19



# QUAD 34 service data

Fig. 19 (continued)



## 6. VOLUME POT

Return all tone and filters to zero and set volume to 21.  
Reset oscilloscope timebase and sensitivity and set the signal generator to apply 10kHz sine waves at 100mV Rms.  
Observe the AC microvoltmeter to ensure that the output levels from both channels are equal, within  $\pm 0.5\text{dB}$ .  
Decrease the volume from 21 to 0 checking the output levels for each setting (listed below) and adjusting the AC microvoltmeter range as necessary. Repeat the test on the other channel.  
The figures below show the attenuation level in dB's with respect to full output for each volume setting. Also quoted are the actual volume circuit amplification and attenuation levels. Tolerance is  $\pm 0.5\text{dB}$ .

| Numerical Setting | Attenuation (w.r.t. full o/p) dB | Amplification (vol. circuit) dB |
|-------------------|----------------------------------|---------------------------------|
| 21                | 0                                | 31.7                            |
| 20                | 2                                | 29.7                            |
| 19                | 4.1                              | 27.6                            |
| 18                | 6.3                              | 25.4                            |
| 17                | 8.5                              | 23.2                            |
| 16                | 10.8                             | 20.9                            |
| 15                | 13.2                             | 18.5                            |
| 14                | 15.6                             | 16.1                            |
| 13                | 18.0                             | 13.7                            |
| 12                | 20.5                             | 11.2                            |
| 11                | 22.9                             | 8.8                             |
| 10                | 25.4                             | 6.3                             |
| 9                 | 27.8                             | 3.9                             |
| 8                 | 30.3                             | 1.4                             |
| 7                 | 32.8                             | -1.1                            |
| 6                 | 35.3                             | -3.6                            |
| 5                 | 37.8                             | -6.1                            |
| 4                 | 42.8                             | -11.1                           |
| 3                 | 47.8                             | -16.1                           |
| 2                 | 52.8                             | -21.1                           |
| 1                 | 57.8                             | -26.1                           |
| 0                 | 89.0                             | -00                             |

Return the volume to 21 and adjust the microvoltmeter accordingly.

7. CROSSTALK

Remove the signal input lead and cover. Solder a 1K resistor across the Radio right channel input. Ensure that S1 is set to drive the left channel then re-connect the leads. Note the output level in dB's. Monitor the output from the right channel which should be at least 50dB below full output. Remove the signal input lead and solder the 1K resistor across the Radio left channel input. Repeat the test driving the right channel and monitoring the output from the left channel to obtain the same results.

N.B. If test equipment leads are not screened or correctly routed it may be difficult to achieve this figure.

8. NOISE

Remove the 1KΩ resistor and all leads. Fit the cover on to the 34. Plug in the AC supply lead, the output lead and the 1K DIN plug dummy loads into Radio, Aux/CD and Tape. Connect a suitably screened approx. 1–10mV cartridge to the 3mV disc input to represent the correct loading. If a moving coil cartridge is being tested use the s/c phono plugs to represent the source impedance.

With the volume control at zero the output level with any selected input should be at least –90dB with respect to full output. If 'A' weighting is used this figure should exceed –100dB.

With the volume at 21 the output levels should be as shown below, again with respect to full output.

| INPUT                | 'A' weighted S/N ratio (dB) | Flat S/N ratio (dB) |
|----------------------|-----------------------------|---------------------|
| Radio                | 88                          | 80                  |
| Auxiliary (pre-8000) | 88                          | 80                  |
| CD/Aux (post 8000)   | 87                          | 79                  |
| Tape                 | 87                          | 79                  |
| Disc 3mV             | 75                          | 65                  |
| 100μV                | 68                          | 53                  |
| 200μV                | 72                          | 57                  |

With Disc selected connect an amplifier and headphones/loudspeaker system and listen to the noise. Randomly switch all tone and filters listening for electrical clicks. Also switch randomly all input buttons again listening for clicks. Clicks due to dirty switch contacts can be cured by squirting the faulty switch with switch cleaner such as WD40.

# QUAD 34 service data

## modifications

### **Approx Serial Number 2000**

ISS 2 p.c.b. introduced incorporating the following changes;

1. Wire links replaced by printed copper links on the top side of the p.c.b.
2. Two holes added in the power supply area of the p.c.b. These are to house two electrically redundant tags added to the transformer to ensure that the transformer is mechanically secured.

### **Approx Serial Number 3700 FEBRUARY 1983**

R55 and R60 changed from 680K to 2M2, stock number R2M20J1. This gives improved bass response.

### **Approx Serial Number 4000 MARCH 1983**

C83 added to decouple the HT rails to earth. C83 is soldered to the rear side of the p.c.b. and on some 34's is 47nf, stock number C47N0ZL soldered between negative rail and earth whilst on other units C83 is 680nf, stock number C680NKT soldered between positive rail and earth.

### **Serial Number 4130 MARCH 1983**

Red filter 1, filter 2 and slope buttons replaced by similar brown buttons, stock numbers MB00F1B, MB00F2B and MB00SLB respectively.

### **Serial Number 6001 JUNE 1983**

ISS 4 p.c.b. introduced incorporating the following changes;

1. Radio analogue switch control is now derived directly from pin 10 of IC11. This reduces selection clicks.
2. C81 and C82, 47n stock number C47N0ZL added to minimise the effects of external RF clicks.
3. R125, 220 $\Omega$  stock number R220RG1 added to protect zener diode D32.
4. C44 moved from the input selector board to the main p.c.b. whilst keeping the same electrical location.

### **Serial Number 8001 NOVEMBER 1983**

ISS 5 p.c.b. introduced incorporating the following changes;

1. C83 removed. C84, 22 $\mu$  stock number C22U0ZE fitted to the top side of the board between negative rail and earth. C58 is moved to be connected between positive rail and earth.
2. Tape and Auxiliary input circuits completely changed. Auxiliary input sensitivity is now 300mV or 100mV according to the plug in flag selection. With 300mV sensitivity auxiliary is suitable for use with compact disc.

### **Serial Number 8305 DECEMBER 1983**

34's are now fitted with CD select buttons stock number MB00CDY, and chassis print is changed accordingly.

complete parts list

| Component Ref. | Value | Stock No. | Previous Value | Previous Stock No. | Comments                   |
|----------------|-------|-----------|----------------|--------------------|----------------------------|
| R3             | 560R  | R560RJ1   |                |                    |                            |
| R4             | 470R  | R470RJ1   |                |                    |                            |
| R7             | 470R  | R470RJ1   |                |                    |                            |
| R8             | 680R  | R680RJ1   |                |                    |                            |
| R11            | 100K  | R100KJ1   |                |                    |                            |
| R12            |       |           | 100K           | R100KJ1            | Removed at ISS 5           |
| R13            | 100K  | R100KJ1   |                |                    |                            |
| R14            |       |           | 100K           | R100KJ1            | Removed at ISS 5           |
| R15            | 2K2   | R2K20G1   |                |                    |                            |
| R16            |       |           | FLAG           |                    | Removed at ISS 5           |
| R17            | 2K2   | R2K20G1   |                |                    |                            |
| R18            |       |           | FLAG           |                    | Removed at ISS 5           |
| R19            |       |           | 39K            | R39K0J1            | Removed at ISS 5           |
| R20            |       |           | 39K            | R39K0J1            | Removed at ISS 5           |
| R25            | 2K2   | R2K20G1   |                |                    |                            |
| R26            | 2K2   | R2K20G1   |                |                    |                            |
| R27            | FLAG  | Q34300R   |                |                    |                            |
| R28            | FLAG  | Q34300R   |                |                    |                            |
| R29            | 10M   | R10M0KB   |                |                    |                            |
| R30            | 10M   | R10M0KB   |                |                    |                            |
| R31            | 10M   | R10M0KB   |                |                    |                            |
| R32            | 10M   | R10M0KB   |                |                    |                            |
| R33            | 4K7   | R4K70J1   |                |                    |                            |
| R34            | 54K9  | R54K9FN   |                |                    |                            |
| R35            | 4K99  | R4K99FN   |                |                    |                            |
| R36            | 750K  | R750KG1   |                |                    |                            |
| R37            | 4K7   | R4K70J1   |                |                    |                            |
| R38            | 54K9  | R54K9FN   |                |                    |                            |
| R39            | 4K99  | R4K99FN   |                |                    |                            |
| R40            | 750K  | R750KG1   |                |                    |                            |
| R41            | 750R  | R750RFN   |                |                    |                            |
| R42            | 39K   | R39K0J1   |                |                    |                            |
| R43            | 750R  | R750RFN   |                |                    |                            |
| R44            | 39K   | R39K0J1   |                |                    |                            |
| R45            | 1K3   | R1K30G1   |                |                    |                            |
| R46            | 1K3   | R1K30G1   |                |                    |                            |
| R47            | 100K  | R100KJ1   |                |                    |                            |
| R48            | 100K  | R100KJ1   |                |                    |                            |
| R49            | 10K   | R10K0J1   |                |                    |                            |
| R50            | 1M    | R1M00J1   |                |                    |                            |
| R51            | 91K   | R91K0G1   |                |                    |                            |
| R52            | 91K   | R91K0G1   |                |                    |                            |
| R53            | 56K   | R56K0G1   |                |                    |                            |
| R54            | 820K  | R820KJ1   |                |                    |                            |
| R55            | 2M2   | R2M20J1   | 680K           | R680KJ1            | Changed at approx S/N 3700 |
| R56            | 91K   | R91K0G1   |                |                    |                            |
| R57            | 91K   | R91K0G1   |                |                    |                            |
| R58            | 56K   | R56K0G1   |                |                    |                            |
| R59            | 820K  | R820KJ1   |                |                    |                            |
| R60            | 2M2   | R2M20J1   | 680K           | R680KJ1            | Changed at approx S/N 3700 |
| R61            | 10K   | R10K0J1   |                |                    |                            |
| R62            | 1M    | R1M00J1   |                |                    |                            |
| R63            | 91K   | R91K0G1   |                |                    |                            |
| R64            | 91K   | R91K0G1   |                |                    |                            |
| R65            | 56K   | R56K0G1   |                |                    |                            |
| R66            | 820K  | R820KJ1   |                |                    |                            |
| R67            | 91K   | R91K0G1   |                |                    |                            |
| R68            | 91K   | R91K0G1   |                |                    |                            |
| R69            | 56K   | R56K0G1   |                |                    |                            |
| R70            | 820K  | R820KJ1   |                |                    |                            |
| R71            | 10K   | R10K0J1   |                |                    |                            |
| R72            | 1M    | R1M00J1   |                |                    |                            |
| R73            | 15K   | R15K0J1   |                |                    |                            |
| R74            | 8K2   | R8K20J1   |                |                    |                            |
| R75            | 18K   | R18K0J1   |                |                    |                            |
| R76            | 39K   | R39K0J1   |                |                    |                            |
| R77            | 15K   | R15K0J1   |                |                    |                            |
| R78            | 8K2   | R8K20J1   |                |                    |                            |
| R79            | 18K   | R18K0J1   |                |                    |                            |
| R80            | 39K   | R39K0J1   |                |                    |                            |
| R81            | 10M   | R10M0KB   |                |                    |                            |
| R82            | 1M    | R1M00J1   |                |                    |                            |
| R83            | 1M    | R1M00J1   |                |                    |                            |
| R84            | 180R  | R180RJ1   |                |                    |                            |
| R85            | 300R  | R300RJ1   |                |                    |                            |

QUAD 34 service data

| Component Ref. | Value                  | Stock No. | Previous Value | Previous Stock No. | Comments         |
|----------------|------------------------|-----------|----------------|--------------------|------------------|
| R86            | 2M2                    | R2M20J1   |                |                    |                  |
| R87            | 2M2                    | R2M20J1   |                |                    |                  |
| R88            | 10M                    | R10M0KB   |                |                    |                  |
| R89            | 1M                     | R1M00J1   |                |                    |                  |
| R90            | 1M                     | R1M00J1   |                |                    |                  |
| R91            | 10K                    | R10K0G1   |                |                    |                  |
| R92            | 22K                    | R22K0J1   |                |                    |                  |
| R93            | 10K                    | R10K0G1   |                |                    |                  |
| R94            | 22K                    | R22K0J1   |                |                    |                  |
| R95            | 1K                     | R1K00J1   |                |                    |                  |
| R96            | 1K                     | R1K00J1   |                |                    |                  |
| R97            | 1K                     | R1K00J1   |                |                    |                  |
| R98            | 10M                    | R10M0KB   |                |                    |                  |
| R100           | 1M                     | R1M00J1   |                |                    |                  |
| R101           | 33K                    | R33K0J1   |                |                    |                  |
| R102           | 22K                    | R22K0J1   |                |                    |                  |
| R103           | 3K3                    | R3K30J1   |                |                    |                  |
| R104           | 3K3                    | R3K30J1   |                |                    |                  |
| R105           | 2M2                    | R2M20J1   |                |                    |                  |
| R106           | 2M2                    | R2M20J1   |                |                    |                  |
| R107           | 10M                    | R10M0KB   |                |                    |                  |
| R108           | 1M                     | R1M00J1   |                |                    |                  |
| R109           | 3K3                    | R3K30J1   |                |                    |                  |
| R110           | 33K                    | R33K0J1   |                |                    |                  |
| R111           | 8K2                    | R8K20J1   |                |                    |                  |
| R112           | 8K2                    | R8K20J1   |                |                    |                  |
| R113           | 10M                    | R10M0KB   |                |                    |                  |
| R114           | 1M                     | R1M00J1   |                |                    |                  |
| R115           | 330K                   | R330KJ1   |                |                    |                  |
| R116           | 3K3                    | R3K30J1   |                |                    |                  |
| R117           | 2K7                    | R2K70J1   |                |                    |                  |
| R118           | 2K2                    | R2K20G1   |                |                    |                  |
| R119           | 1K                     | R1K00G1   |                |                    |                  |
| R120           | 2K7                    | R2K70J1   |                |                    |                  |
| R121           | 2K2                    | R2K20G1   |                |                    |                  |
| R122           | 1K                     | R1K00G1   |                |                    |                  |
| R123           | 8K2                    | R8K20J1   |                |                    |                  |
| R124           | 9K1                    | R9K10J1   |                |                    |                  |
| R125           | 220R                   | R220RG1   |                |                    | Added at ISS 4   |
| R131           | 39K                    | R39K0J1   |                |                    | Added at ISS 5   |
| R132           | 39K                    | R39K0J1   |                |                    | Added at ISS 5   |
| R133           | 39K                    | R39K0J1   |                |                    | Added at ISS 5   |
| R134           | 39K                    | R39K0J1   |                |                    | Added at ISS 5   |
| R135           | FLAG                   | Q34V5CD   |                |                    | Added at ISS 5   |
| R136           | FLAG                   | Q34V5CD   |                |                    | Added at ISS 5   |
| R137           | FLAG                   | Q34300P   |                |                    | Added at ISS 5   |
| R138           | FLAG                   | Q34300P   |                |                    | Added at ISS 5   |
| R139           | 1M                     | R1M00J4   |                |                    | Added at ISS 5   |
| R140           | 1M                     | R1M00J4   |                |                    | Added at ISS 5   |
| R141           | 1M                     | R1M00J4   |                |                    | Added at ISS 5   |
| R142           | 1M                     | R1M00J4   |                |                    | Added at ISS 5   |
| R143           | 10K                    | R10K0J4   |                |                    | Added at ISS 5   |
| R144           | 10K                    | R10K0J4   |                |                    | Added at ISS 5   |
| R145           | 10K                    | R10K0J4   |                |                    | Added at ISS 5   |
| R146           | 10K                    | R10K0J4   |                |                    | Added at ISS 5   |
| X1             | Flag Record Link to CD | Q34X1X2   |                |                    |                  |
| X2             | Flag Record Link to CD | Q34X1X2   |                |                    |                  |
| SF1            | Flag Tape 100mV Replay | Q34100P   |                |                    |                  |
| SF2            | Flag Tape 100mV Replay | Q34100P   |                |                    |                  |
| SF3            | Flag Tape 100mV Record | Q34100R   |                |                    |                  |
| SF4            | Flag Tape 100mV Record | Q34100R   |                |                    |                  |
| SF5            | Flag Tape 300mV Replay | Q34300P   |                |                    |                  |
| SF6            | Flag Tape 300mV Replay | Q34300P   |                |                    |                  |
| SF7            | Flag Tape 300mV Record | Q34300R   |                |                    |                  |
| SF8            | Flag Tape 300mV Record | Q34300R   |                |                    |                  |
| SF9            | Flag CD 500mV          | Q34V5CD   |                |                    |                  |
| SF10           | Flag CD 500mV          | Q34V5CD   |                |                    |                  |
| D1             | 5V6 Zener              | D795V6A   |                |                    |                  |
| D2             | 5V6 Zener              | D795V6A   |                |                    |                  |
| D3             | 6V8 Zener              | D886V8A   |                |                    |                  |
| D4             | 6V8 Zener              | D886V8A   |                |                    |                  |
| D5             | 6V8 Zener              | D886V8A   |                |                    |                  |
| D6             | 6V8 Zener              | D886V8A   |                |                    |                  |
| D7             |                        |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |
| D8             |                        |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |
| D9             |                        |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |
| D10            |                        |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |
| D11            |                        |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |

QUAD 34 service data

| Component Ref. | Value     | Stock No. | Previous Value | Previous Stock No. | Comments         |
|----------------|-----------|-----------|----------------|--------------------|------------------|
| D12            |           |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |
| D13            |           |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |
| D14            |           |           | 6V8 Zener      | D886V8A            | Removed at ISS 5 |
| D15            | TLG124    | BLG124P   |                |                    |                  |
| D16            | TLG124    | BLG124P   |                |                    |                  |
| D17            | TLG124    | BLG124P   |                |                    |                  |
| D18            | TLG124    | BLG124P   |                |                    |                  |
| D19            | TLG124    | BLG124P   |                |                    |                  |
| D20            | TLR124    | BLR124P   |                |                    |                  |
| D21            | TLR124    | BLR124P   |                |                    |                  |
| D22            | TLR124    | BLR124P   |                |                    |                  |
| D23            | IN4148    | D1N4148   |                |                    |                  |
| D24            | IN4148    | D1N4148   |                |                    |                  |
| D25            | IN4148    | D1N4148   |                |                    |                  |
| D26            | IN4148    | D1N4148   |                |                    |                  |
| D27            | IN4148    | D1N4148   |                |                    |                  |
| D28            | IN4148    | D1N4148   |                |                    |                  |
| D29            | IN4148    | D1N4148   |                |                    |                  |
| D30            | 15V Zener | D8815VA   |                |                    |                  |
| D31            | IN4148    | D1N4148   |                |                    |                  |
| D32            | 12V Zener | D8812VA   |                |                    |                  |
| D33            | TLG124    | BLG124T   |                |                    |                  |
| D34            | VM18      | DVM18XX   |                |                    |                  |
| C3             | 330n      | C330NKS   |                |                    |                  |
| C4             | 330n      | C330NKS   |                |                    |                  |
| C5             |           |           | 330n           | C330NKS            | Removed at ISS 5 |
| C6             |           |           | 330n           | C330NKS            | Removed at ISS 5 |
| C7             |           |           | 680n           | C680NKS            | Removed at ISS 5 |
| C8             |           |           | 680n           | C680NKS            | Removed at ISS 5 |
| C10            | 100μ      | C100UME   |                |                    |                  |
| C11            | 100μ      | C100UME   |                |                    |                  |
| C16            | 100μ      | C100UME   |                |                    |                  |
| C17            | 100μ      | C100UME   |                |                    |                  |
| C18            | 2μ2       | C2U20KJ   |                |                    |                  |
| C19            | 47n       | C47N0FA   |                |                    |                  |
| C20            | 15n       | C15N0FA   |                |                    |                  |
| C21            | 47p       | C47P0KJ   |                |                    |                  |
| C22            | 2μ2       | C2U20KJ   |                |                    |                  |
| C23            | 47n       | C47N0FA   |                |                    |                  |
| C24            | 15n       | C15N0FA   |                |                    |                  |
| C25            | 47p       | C47P0KJ   |                |                    |                  |
| C26            | 15n       | C15N0FA   |                |                    |                  |
| C27            | 470n      | C470NKS   |                |                    |                  |
| C28            | 15n       | C15N0FA   |                |                    |                  |
| C29            | 470n      | C470NKS   |                |                    |                  |
| C30            | 100μ      | C100UME   |                |                    |                  |
| C31            | 100μ      | C100UME   |                |                    |                  |
| C32            | 1n        | C1N00JP   |                |                    |                  |
| C33            | 1n        | C1N00JP   |                |                    |                  |
| C34            | 15p       | C15P0JI   |                |                    |                  |
| C35            | 15p       | C15P0JI   |                |                    |                  |
| C36            | 220n      | C220NKS   |                |                    |                  |
| C37            | 220n      | C220NKS   |                |                    |                  |
| C38            | 220n      | C220NKS   |                |                    |                  |
| C39            | 220n      | C220NKS   |                |                    |                  |
| C40            | 3n3       | C3N30JP   |                |                    |                  |
| C41            | 1n        | C1N00JP   |                |                    |                  |
| C42            | 3n3       | C3N30JP   |                |                    |                  |
| C43            | 1n        | C1N00JP   |                |                    |                  |
| C44            | 4n7       | C4N70ZM   |                |                    |                  |
| C45            | 3n3       | C3N30JP   |                |                    |                  |
| C46            | 1n        | C1N00JP   |                |                    |                  |
| C47            | 220n      | C220NKS   |                |                    |                  |
| C48            | 100μ      | C100UKT   |                |                    |                  |
| C49            | 3n3       | C3N30JP   |                |                    |                  |
| C50            | 1n        | C1N00JP   |                |                    |                  |
| C51            | 68n       | C68N0KS   |                |                    |                  |
| C52            | 68n       | C68N0KS   |                |                    |                  |
| C53            | 220n      | C220NKS   |                |                    |                  |
| C54            | 100μ      | C100UKT   |                |                    |                  |
| C55            | 47n       | C47N0ZL   |                |                    |                  |
| C56            | 4n7       | C4N70ZM   |                |                    |                  |
| C57            | 100μ      | C100UME   |                |                    |                  |
| C58            | 22μ       | C22U0ZE   |                |                    |                  |
| C59            | 100μ      | C100UME   |                |                    |                  |
| C61            | 680p      | C680PKJ   |                |                    |                  |
| C62            | 1n        | C1N00JP   |                |                    |                  |
| C63            | 680p      | C680PKJ   |                |                    |                  |
| C64            | 1n        | C1N00JP   |                |                    |                  |
| C65            | 47n       | C47N0ZL   |                |                    |                  |
| C66            | 4n7       | C4N70ZM   |                |                    |                  |

# QUAD 34 service data

| Component Ref. | Value                       | Stock No. | Previous Value | Previous Stock No. | Comments          |
|----------------|-----------------------------|-----------|----------------|--------------------|-------------------|
| C67            | 47n                         | C47N0ZL   |                |                    |                   |
| C68            | 4n7                         | C4N70ZM   |                |                    |                   |
| C69            | 100μ                        | C100UME   |                |                    |                   |
| C70            | 1n                          | C1N00JP   |                |                    |                   |
| C71            | 220p                        | C220PJJ   |                |                    |                   |
| C72            | 1n                          | C1N00JP   |                |                    |                   |
| C73            | 220p                        | C220PJJ   |                |                    |                   |
| C74            | 1000μ                       | C1K0UTD   |                |                    |                   |
| C75            | 120p                        | C120PJP   |                |                    |                   |
| C76            | 120p                        | C120PJP   |                |                    |                   |
| C77            | 100μ                        | C100UME   |                |                    |                   |
| C78            | 100μ                        | C100UME   |                |                    |                   |
| C79            | 4n7                         | C4N70ZM   |                |                    |                   |
| C80            | 4n7                         | C4N70ZM   |                |                    |                   |
| C81            | 47n                         | C47N0ZL   |                |                    | Added at ISS 4    |
| C82            | 47n                         | C47N0ZL   |                |                    | Added at ISS 4    |
| C83            |                             |           |                |                    | See modifications |
| C84            | 22μ                         | C22U0ZE   |                |                    | Added at ISS 5    |
| C85            | 680n                        | C680NKA   |                |                    | Added at ISS 5    |
| C86            | 680n                        | C680NKA   |                |                    | Added at ISS 5    |
| C87            | 680n                        | C680NKA   |                |                    | Added at ISS 5    |
| C88            | 680n                        | C680NKA   |                |                    | Added at ISS 5    |
| C89            | 100μ                        | C100UME   |                |                    | Added at ISS 5    |
| C90            | 100μ                        | C100UME   |                |                    | Added at ISS 5    |
| C91            | 100μ                        | C100UME   |                |                    | Added at ISS 5    |
| C92            | 100μ                        | C100UME   |                |                    |                   |
| IC1            | 4016 Quad Transmission Gate | DCD4016   |                |                    |                   |
| IC2            | 4066 Quad Transmission Gate | DCD4066   |                |                    |                   |
| IC3            |                             |           | 071 OP AMP     | D071CPX            | Removed at ISS 5  |
| IC4            |                             |           | 071 OP AMP     | D071CPX            | Removed at ISS 5  |
| IC5            |                             |           | 071 OP AMP     | D071CPX            | Removed at ISS 5  |
| IC6            |                             |           | 071 OP AMP     | D071CPX            | Removed at ISS 5  |
| IC7            | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC8            | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC9            | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC10           | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC11           | 4001 Quad Two Input NOR     | DCD4001   |                |                    |                   |
| IC12           | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC13           | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC14           | 4013 Dual D-Type Latch      | DCD4013   |                |                    |                   |
| IC15           | 4013 Dual D-Type Latch      | DCD4013   |                |                    |                   |
| IC16           | 4013 Dual D-Type Latch      | DCD4013   |                |                    |                   |
| IC17           | 4066 Quad Transmission Gate | DCD4066   |                |                    |                   |
| IC18           | 4066 Quad Transmission Gate | DCD4066   |                |                    |                   |
| IC19           | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC20           | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC21           | 7918 18V Regulator          | DUA7918   |                |                    |                   |
| IC22           | 4066 Quad Transmission Gate | DCD4066   |                |                    |                   |
| IC23           | 071 Op Amp                  | D071CPX   |                |                    |                   |
| IC24           | 072 Dual 071                | D072CPX   |                |                    | Added at ISS 5    |
| IC25           | 072 Dual 071                | D072CPX   |                |                    | Added at ISS 5    |
| IC26           | 072 Dual 071                | D072CPX   |                |                    | Added at ISS 5    |
| IC27           | 072 Dual 071                | D072CPX   |                |                    | Added at ISS 5    |
| T5             | BC183                       | DBC183L   |                |                    |                   |
| T6             | BC183                       | DBC183L   |                |                    |                   |
| T7             | BC183                       | DBC183L   |                |                    |                   |
| T8             | BC183                       | DBC183L   |                |                    |                   |
| T9             | BC183                       | DBC183L   |                |                    |                   |
| T10            | BC183                       | DBC183L   |                |                    |                   |
| T11            | BC183                       | DBC183L   |                |                    |                   |
| T12            | BC183                       | DBC183L   |                |                    |                   |
| T13            | BC413                       | DBC413X   |                |                    |                   |
| T14            | BC413                       | DBC413X   |                |                    |                   |
| T15            | BC413                       | DBC413X   |                |                    |                   |
| S1             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S2             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S3             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S4             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S5             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S6             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S7             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S8             | Input Selector Switch       | S44INPA   |                |                    |                   |
| S9             | Voltage Selector Switch     | SVL1869   |                |                    |                   |
| S10            | Mains Switch                | SF4OFFA   |                |                    |                   |
| L1             | Mains Transformer           | M12727A   |                |                    |                   |

| Component |                                           | Stock No. | Comments                                       |
|-----------|-------------------------------------------|-----------|------------------------------------------------|
| Ref.      |                                           |           |                                                |
| RV1       | Volume Pot                                | R34VOLA   |                                                |
| RV2       | Tilt Pot                                  | R34TILA   |                                                |
| RV3       | Bass Pot                                  | R34BASA   |                                                |
| RV4       | Balance Pot                               | R34BALB   |                                                |
| SK3       | 5 Pin Din SKT                             | PS05DNB   |                                                |
| SK4       | 5 Pin Din SKT                             | PS05DNB   |                                                |
| SK5       | 5 Pin Din SKT                             | PS05DNB   |                                                |
| SK6       | Mains Outlet                              | PSP695S   |                                                |
| SK7       | 4 Pin Din SKT                             | PS04DNB   |                                                |
| PL1       | Mains in SKT                              | PPR0331   |                                                |
| N1        | Suppressor                                | NPMR20A   |                                                |
| FS1       | 63 Ma Fuse                                | UM63MDA   |                                                |
| MISC      |                                           |           |                                                |
| 1         | Front Panel                               | M12560P   |                                                |
| 2         | Chassis                                   | M12567A   |                                                |
| 3         | Cover                                     | M12573A   |                                                |
| 4         | Sub-Panel                                 | M12574A   |                                                |
| 5         | Front Plate                               | M12570A   |                                                |
| 6         | Fuse Holder                               | PF5234A   |                                                |
| 7         | Transformer Screen                        | M12571A   |                                                |
| 8         | Transformer Cover                         | M12572A   |                                                |
| 9         | Safety Cover                              | M12575A   |                                                |
| 10        | Voltage Selector Shroud                   | M12579A   |                                                |
| 11        | Mains Switch Link                         | M12771A   | Replaces 3 Part Switch Link used on early 34's |
| 12        | PC Board Main                             | I12730A   |                                                |
| 13        | PC Board Button                           | I12729A   |                                                |
| 14        | Balance Control Knob                      | M12578K   |                                                |
| 15        | Balance Control Lever                     | M12578L   |                                                |
| 16        | Tone Control Knob                         | M12450P   |                                                |
| 17        | Earth Terminal                            | PQ1124A   |                                                |
| 18        | Cover Guide                               | M12562A   |                                                |
| 19        | Foot Moulding                             | M12620A   |                                                |
| 20        | Non-Slip Foot                             | AFNOSLA   |                                                |
| 21        | Heatsink T0220                            | NHSTV5A   |                                                |
| 22        | Wire Link for Balance Control             | M12587A   |                                                |
| 23        | 8-Way Cable                               | M127118   |                                                |
| 24        | 10-Way Cable                              | M12723A   |                                                |
| 25        | M2.5 x 6mm Screws Black                   | TM206PB   |                                                |
| 26        | M3 x 6mm Screws                           | TM306PA   |                                                |
| 27        | M2.5 x 5mm Screws                         | TM205PA   |                                                |
| 28        | Screw to hold Transformer Cover           | TC406PC   |                                                |
| 29        | Screw to hold Din Sockets to Back         | TC205PF   |                                                |
| 30        | M3 x 6mm Stainless Steel                  | TM306PG   |                                                |
| 31        | Grub Screw M3 x 10mm                      | TM310GC   |                                                |
| 32        | Grub Screw M3 x 16mm Vol. Control         | TM316GC   |                                                |
| 33        | M4 x 16mm Cover Fixing                    | TM416PA   |                                                |
| 34        | M5 x 5mm Foot Screw                       | TM405PA   |                                                |
| 35        | Serial Number Label                       | M12734A   |                                                |
| 36        | Push On Fix for Sub-Plate                 | FF123ZF   |                                                |
| 37        | Mains Input Socket Cable Mounting         | PSR0113   |                                                |
| 38        | Mains Interconnecting Lead                | QSPES1A   |                                                |
| 39        | 5 Pin Din to 4 Phono Lead                 | QD5FP4A   |                                                |
| 40        | 4 Pin Din to 4 Pin Din Lead               | QD4D41A   |                                                |
| 41        | Sound Deadening                           | IPEDAMA   |                                                |
| 42        | Volume Knob                               | M12586A   |                                                |
| 43        | Led Bezel                                 | M12710A   |                                                |
| 44        | Molex Jumper Assy for Disc                | M12724A   |                                                |
| 45        | Sockets for Plug In Coïmps                | PAM5982   |                                                |
| 46        | W Button to fix Safety Cover              | FP70271   |                                                |
| 47        | Warning Label                             | M12663A   |                                                |
| 48        | Clip for Balance Control Linkage          | FF3419A   |                                                |
| 49        | Wire Type 3 Brown                         | WM80131   |                                                |
| 50        | Wire Type 3 Blue                          | WM80136   |                                                |
| 51        | Wire Type 3 Green/Yellow                  | WM8013E   |                                                |
| 52        | Washer Shake Proof for Signal Earth Screw | TDB4NLF   |                                                |
| 53        | Washer behind Bal. Sleeve                 | M12759A   |                                                |
| 54        | Washers for Bal. Control M2.5 Plain       | TDB8NPA   |                                                |
| 55        | Washers for Cover Screws M4 Plain         | TDM4SPA   |                                                |
| 56        | Nut for Regulator Heat Sink M3            | TM3FHPA   |                                                |
| 57        | Washer M3 x 1mm Thick                     | TDB6NPA   |                                                |
| 58        | Sil Rubber Sleeving for Bal. Link         | ISFIMS2   |                                                |
| 59        | Nut for Mains Earth Fixing M3             | TM3FHPA   |                                                |
| 60        | Shakeproof Washer for Mains Earth Fixing  | TDB6NLF   |                                                |
| 61        | Soldertag for Mains Earth Fixing          | FTB6SS5   |                                                |
| 62        | Label for QDEFP4A                         | M12751A   |                                                |

# QUAD 34 service data

| Component Ref.                         |                                  | Stock No. | Previous Stock No. | Comments            |
|----------------------------------------|----------------------------------|-----------|--------------------|---------------------|
| 63                                     | RAD Button                       | MBRADIY   |                    |                     |
| 64                                     | Disc Button                      | MBDISCY   |                    |                     |
| 65                                     | Aux Button                       | MB0AUXY   |                    |                     |
| 66                                     | CD Button                        | MB00CDY   |                    |                     |
| 67                                     | Tape Button                      | MBTAPEY   |                    |                     |
| 68                                     | Mono Button                      | MBMONOY   |                    |                     |
| 69                                     | F1 Button                        | MB00F1B   | MB00F1R            | Changed at S/N 4130 |
| 70                                     | F2 Button                        | MB00F2B   | MB00F2R            | Changed at S/N 4130 |
| 71                                     | SL Button                        | MB00SLB   | MB00SLR            | Changed at S/N 4130 |
| MOVING COIL INPUT MODULE 100 MICROVOLT |                                  |           |                    |                     |
| R1a                                    | 100R                             | R100RJ1   |                    |                     |
| R2a                                    | 100R                             | R100RJ1   |                    |                     |
| R5a                                    | 2K2                              | R2K20G1   |                    |                     |
| R6a                                    | 2K2                              | R2K20G1   |                    |                     |
| R9a                                    | 2K2                              | R2K20G1   |                    |                     |
| R10a                                   | 2K2                              | R2K20G1   |                    |                     |
| R21a                                   | 6R8                              | R6R80G1   |                    |                     |
| R22a                                   | 1K1                              | R1K10G1   |                    |                     |
| R23a                                   | 6R8                              | R6R80G1   |                    |                     |
| R24a                                   | 1K1                              | R1K10G1   |                    |                     |
|                                        |                                  |           |                    |                     |
| C1a                                    | 22n                              | C22N0JS   |                    |                     |
| C2a                                    | 22n                              | C22N0JS   |                    |                     |
| C12a                                   | 47μ                              | C47U0KT   |                    |                     |
| C13a                                   | 47μ                              | C47U0KT   |                    |                     |
| C14a                                   | 47μ                              | C47U0KT   |                    |                     |
| C15a                                   | 47μ                              | C47U0KT   |                    |                     |
|                                        |                                  |           |                    |                     |
| T1a                                    | ZTX750                           | DZTX75P   |                    |                     |
| T2a                                    | ZTX650                           | DZTX65P   |                    |                     |
| T3a                                    | ZTX750                           | DZTX75P   |                    |                     |
| T4a                                    | ZTX650                           | DZTX65P   |                    |                     |
|                                        |                                  |           |                    |                     |
| SK1a                                   | Phono Socket                     | PSPHOG9   |                    |                     |
| SK2a                                   | Phono Socket                     | PSPHOG2   |                    |                     |
| MISC                                   |                                  |           |                    |                     |
| 1                                      | PCB                              | I12728A   |                    |                     |
| 2                                      | Insulating Bush for Phono Socket | M12716A   |                    |                     |
| 3                                      | M2.5 x 5mm Screws                | TM205PA   |                    |                     |
| 4                                      | M2.5 Nuts                        | TM2FHPA   |                    |                     |
| 5                                      | Disc Module Front Plate          | M12721A   |                    |                     |
| 6                                      | Disc Module Mounting Bracket     | M12722A   |                    |                     |
| 7                                      | Molex Board Plug                 | PP50468   |                    |                     |
| 8                                      | Packing                          | ZC34DMI   |                    |                     |
| 9                                      | Packing                          | ZX34DMA   |                    |                     |
| 10                                     | Wire for Phono                   | WM60125   |                    |                     |
| 11                                     | Wire for Phono Earth             | WTM71AB   |                    |                     |
| 200 MICROVOLT MOVING COIL INPUT        |                                  |           |                    |                     |
| R1a                                    | 100R                             | R100RJ1   |                    |                     |
| R2a                                    | 100R                             | R100RJ1   |                    |                     |
| R5a                                    | 5K6                              | R5K60G1   |                    |                     |
| R6a                                    | 5K6                              | R5K60G1   |                    |                     |
| R9a                                    | 5K6                              | R5K60G1   |                    |                     |
| R10a                                   | 5K6                              | R5K60G1   |                    |                     |
| R21a                                   | 15R                              | R15R0FN   |                    |                     |
| R22a                                   | 1K1                              | R1K10G1   |                    |                     |
| R23a                                   | 15R                              | R15R0FN   |                    |                     |
| R24a                                   | 1K1                              | R1K10G1   |                    |                     |
|                                        |                                  |           |                    |                     |
| C1a                                    | 22n                              | C22NOJS   |                    |                     |
| C2a                                    | 22n                              | C22NOJS   |                    |                     |
| C12a                                   | 22μ                              | C22U0KB   |                    |                     |
| C13a                                   | 22μ                              | C22U0KB   |                    |                     |
| C14a                                   | 22μ                              | C22U0KB   |                    |                     |
| C15a                                   | 22μ                              | C22U0KB   |                    |                     |
|                                        |                                  |           |                    |                     |
| T1a                                    | ZTX750                           | DZTX75P   |                    |                     |
| T2a                                    | ZTX650                           | DZTX65P   |                    |                     |
| T3a                                    | ZTX750                           | DZTX75P   |                    |                     |
| T4a                                    | ZTX650                           | DZTX65P   |                    |                     |
|                                        |                                  |           |                    |                     |
| SK1a                                   | Phono Socket                     | PSPHOG9   |                    |                     |
| SK2a                                   | Phono Socket                     | PSPHOG2   |                    |                     |

| Component Ref.                         |                                  | Stock No. |
|----------------------------------------|----------------------------------|-----------|
| MISC                                   |                                  |           |
| 1                                      | PC Board                         | I12728A   |
| 2                                      | Insulating Bush for Phono Socket | M12716A   |
| 3                                      | M2.5 x 5mm Screws                | TM205PA   |
| 4                                      | M2.5 Nuts                        | TM2FHPA   |
| 5                                      | Disc Module Front Plate          | M12721A   |
| 6                                      | Disc Module Bracket              | M12722A   |
| 7                                      | Molex Board Plug                 | PP50468   |
| 8                                      | Packing                          | ZC34DMI   |
| 9                                      | Packing                          | ZX34DMA   |
| 10                                     | Wire for Phono                   | WM60125   |
| 11                                     | Wire for Phono Earth             | WTM71AB   |
| 400 Microvolt INPUT                    |                                  |           |
| R1a                                    | 100R                             | R100RJ1   |
| R2a                                    | 100R                             | R100RJ1   |
| R5a                                    | 15K                              | R15KOG1   |
| R6a                                    | 15K                              | R15KOG1   |
| R9a                                    | 15K                              | R15KOG1   |
| R10a                                   | 15K                              | R15KOG1   |
| R21a                                   | 30R                              | R30ROFN   |
| R22a                                   | 1K1                              | R1K1OG1   |
| R23a                                   | 30R                              | R30ROFN   |
| R24a                                   | 1K1                              | R1K1OG1   |
| C1a                                    | 22n                              | C22NOJS   |
| C2a                                    | 22n                              | C22NOJS   |
| C12a                                   | 10μ                              | C10UOKT   |
| C13a                                   | 10μ                              | C10UOKT   |
| C14a                                   | 10μ                              | C10UOKT   |
| C15a                                   | 10μ                              | C10UOKT   |
| T1a                                    | ZTX750                           | DZTX75P   |
| T2a                                    | ZTX650                           | DZTX65P   |
| T3a                                    | ZTX750                           | DZTX75P   |
| T4a                                    | ZTX650                           | DZTX65P   |
| SK1a                                   | Phono Socket                     | PSPHOG9   |
| SK2a                                   | Phono Socket                     | PSPHOG2   |
| MISC                                   |                                  |           |
| 1                                      | PC Board                         | I12728A   |
| 2                                      | Insulating Bush for Phono Socket | M12716A   |
| 3                                      | M2.5 x 5mm Screws                | TM205PA   |
| 4                                      | Nuts M2.5                        | TM2FHPA   |
| 5                                      | Disc Module Front Plate          | M12721A   |
| 6                                      | Disc Module Bracket              | M12722A   |
| 7                                      | Molex Board Plug                 | PP50468   |
| 8                                      | Packing                          | ZC34DMI   |
| 9                                      | Packing                          | ZX34DMA   |
| 10                                     | Wire for Phono                   | WM60125   |
| 11                                     | Wire for Phono Earth             | WTM71AB   |
| MOVING MAGNET INPUT MODULE 3 MILLIVOLT |                                  |           |
| R1b                                    | 47K                              | R47KOJ1   |
| R2b                                    | 47K                              | R47KOJ1   |
| R5b                                    | 82K                              | R82KOJ1   |
| R6b                                    | 82K                              | R82KOJ1   |
| R9b                                    | 82K                              | R82KOJ1   |
| R10b                                   | 82K                              | R82KOJ1   |
| R21b                                   | 220R                             | R220RG1   |
| R22b                                   | 910R                             | R91ORFN   |
| R23b                                   | 220R                             | R220RG1   |
| R24b                                   | 910R                             | R91ORFN   |
| C1b                                    | 220p                             | C220PJJ   |
| C2b                                    | 220p                             | C220PJJ   |
| C12b                                   | 1μ5                              | C1U5OKT   |
| C13b                                   | 1μ5                              | C1U5OKT   |
| C14b                                   | 1μ5                              | C1U5OKT   |
| C15b                                   | 1μ5                              | C1U5OKT   |
| T1b                                    | BC214C                           | DBC214C   |
| T2b                                    | BC413                            | DBC413X   |
| T3b                                    | BC214C                           | DBC214C   |
| T4b                                    | BC413                            | DBC413X   |
| SK1b                                   | Phono Socket                     | PSPHON9   |
| SK2b                                   | Phono Socket                     | PSPHON2   |

# QUAD 34 service data

| Component |                                  | Stock No. |
|-----------|----------------------------------|-----------|
| Ref.      |                                  |           |
| MISC      |                                  |           |
| 1         | PC Board                         | I12728A   |
| 2         | Insulating Bush for Phono Socket | M12716A   |
| 3         | M2.5 x 5mm Screws                | TM205PA   |
| 4         | Nuts M2.5                        | TM2FHPA   |
| 5         | Disc Module Front Plate          | M12721A   |
| 6         | Disc Module Bracket              | M12722A   |
| 7         | Molex Board Plug                 | PP50468   |
| 8         | Wire for Phono                   | WM60125   |
| 9         | Wire for Phono Earth             | WTM71AB   |

# assembly diagram

| No. | Description              | Stock No. |
|-----|--------------------------|-----------|
| 1   | Screw M4 x 16 mm         | TM416PA   |
| 2   | Washer M4                | TDM4SPA   |
| 3   | Chassis Guide            | M12562A   |
| 4   | Cover                    | M12573A   |
| 5   | Foot                     | M12620A   |
| 6   | Screw M4 x 5 mm          | TM405PA   |
| 7   | Non-Slip Pad             | AFNOSLA   |
| 8   | Screw 2M5 x 6 mm         | TM206PB   |
| 9   | Disc Module              |           |
| 10  | 'Earth' Terminal         | PQ1124A   |
| 11  | Screw 2Z x 3/16          | TC205PF   |
| 12  | Fuse T100mA              | UMA10DA   |
| 13  | Fuseholder               | PF5234A   |
| 14  | Voltage Selector Shroud  | M12579A   |
| 15  | Mains Out – Socket       | PSR472A   |
| 16  | Screw 4Z x ¼             | TC406PC   |
| 17  | Mains In – Plug          | PPR0331   |
| 18  | Mains Transformer Cover  | M12572A   |
| 19  | Mains Transformer Screen | M12571A   |
| 20  | Chassis                  | M12567A   |
| 21  | Screw M3 x 6 mm          | TM306PA   |
| 22  | Washer M3                | TDM3SPA   |
| 23  | Volume Control           | R34VOLA   |
| 24  | Tilt Control             | R34TILA   |
| 25  | Bass Control             | R34BASA   |
| 26  | 4 Pin Din Socket         | PSO4DNB   |
| 27  | 5 Pin Din Socket         | PSO5DNB   |
| 28  | Screw M3 x 6 mm          | TM306PA   |
| 29  | 8 Way Cable and Socket   | M12724A   |
| 30  | PCB Assembled            |           |
| 31  | Safety Cover             | M12575A   |
| 32  | 'W' Button               | FP70271   |
| 33  | Screw M3 x 6 mm SS       | TM306PG   |
| 34  | Button Board Assembled   |           |
| 35  | LED Bezel                | M12710A   |
| 36  | Push Button Cap – Set    | M12597A   |
| 37  | Push-On-Fix              | FF123ZF   |
| 38  | Balance Control          | R34BALA   |
| 39  | Front Plate              | M12570A   |
| 40  | Balance Control Lever    | M12578L   |
| 41  | Balance Control Link     | M12587A   |
| 42  | Push-On-Fix              |           |
| 43  | Sub-Panel                | M12574A   |
| 44  | Balance Control Knob     | M12578K   |
| 45  | Mains Switch Link        | M12771A   |
| 46  | Screw 2M5 x 5 mm         | TM205PA   |
| 47  | Washers M2.5             | TDB8NPA   |
| 48  | Grubscrew M3 x 16 mm     | TM316HC   |
| 49  | Volume Control Knob      | M12586A   |
| 50  | Grubscrew M3 x 10 mm     | TM310GC   |
| 51  | Tone Control Knob        | M12450A   |
| 52  | Front Panel              | M12560A   |
| 53  | Washer M3                | TDB6NPA   |
| 54  | Shake Proof Washer 4BA   | TDB4NLF   |
| 55  | Soldertag                | FTB6SS5   |
| 56  | Nut M3                   | TM3FHPA   |
| 57  | Warning Label            | M12663A   |
| 58  | Washer                   | M12759A   |

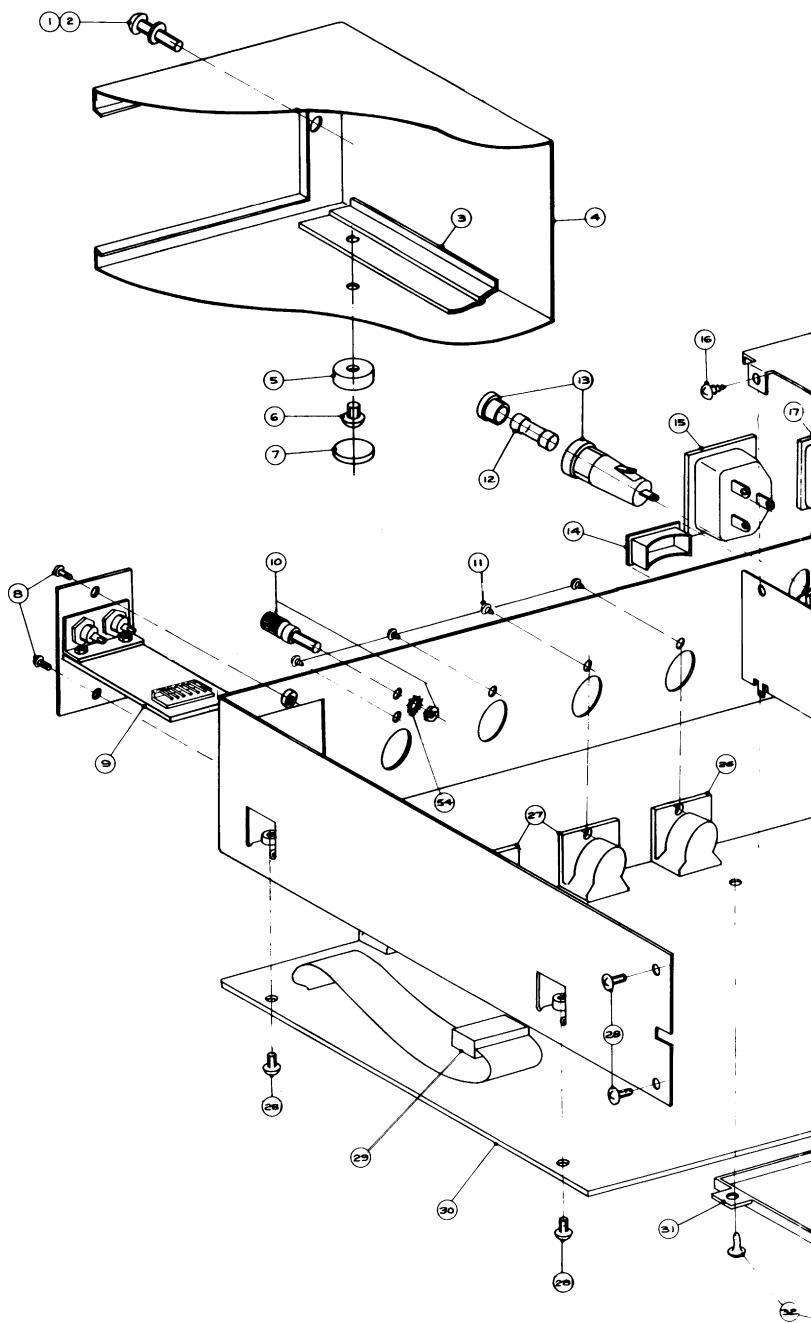
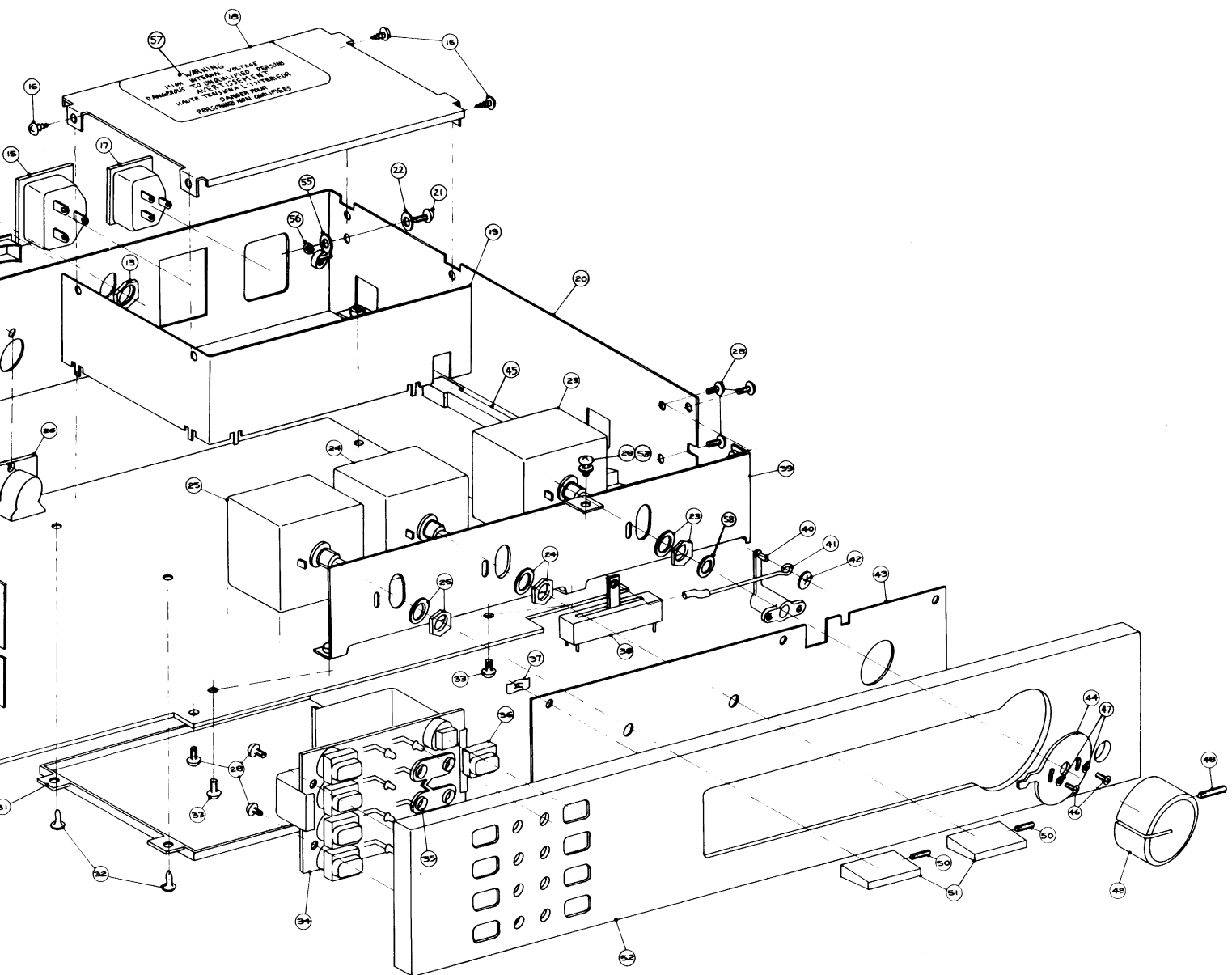


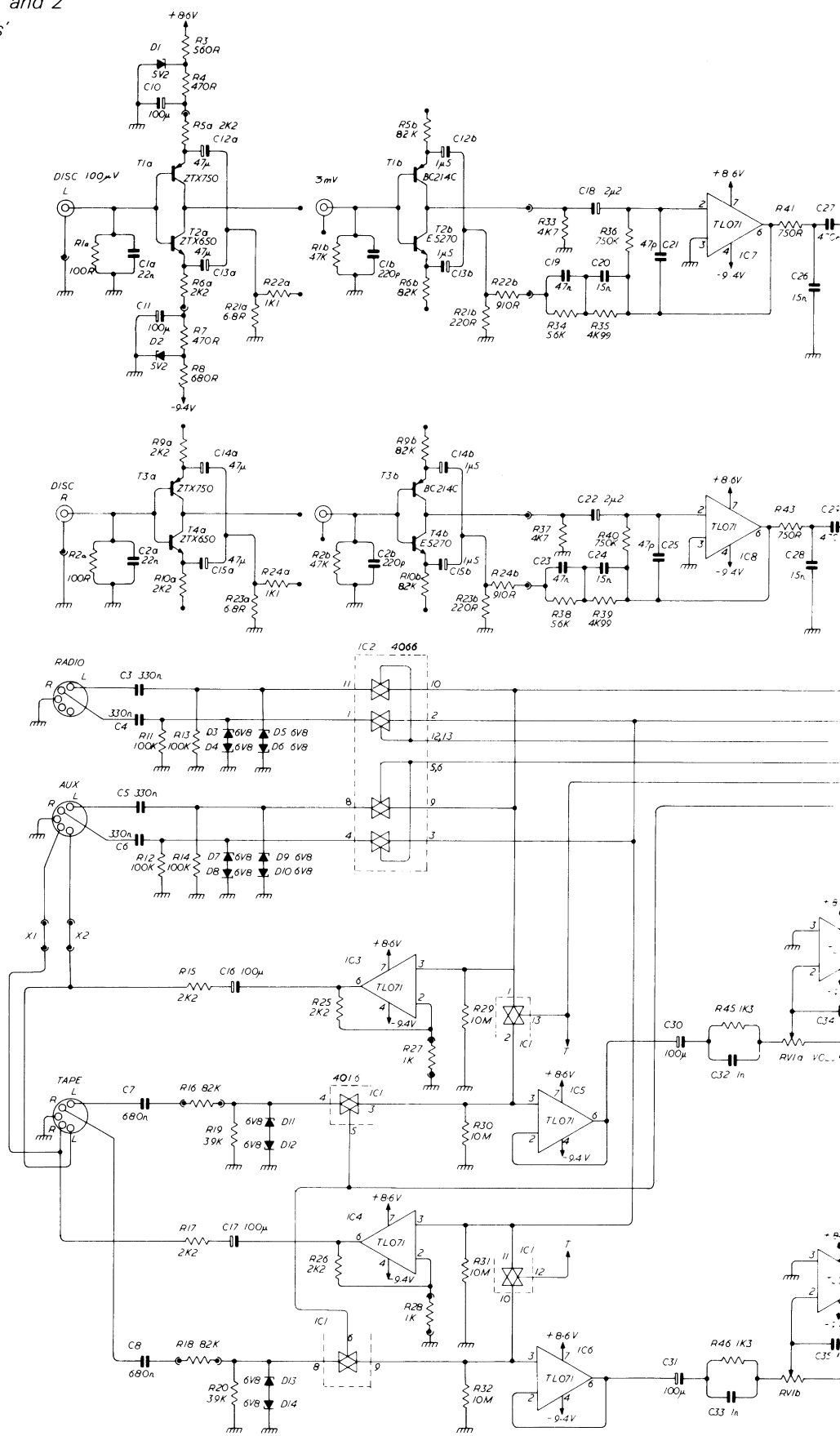
Fig. 20



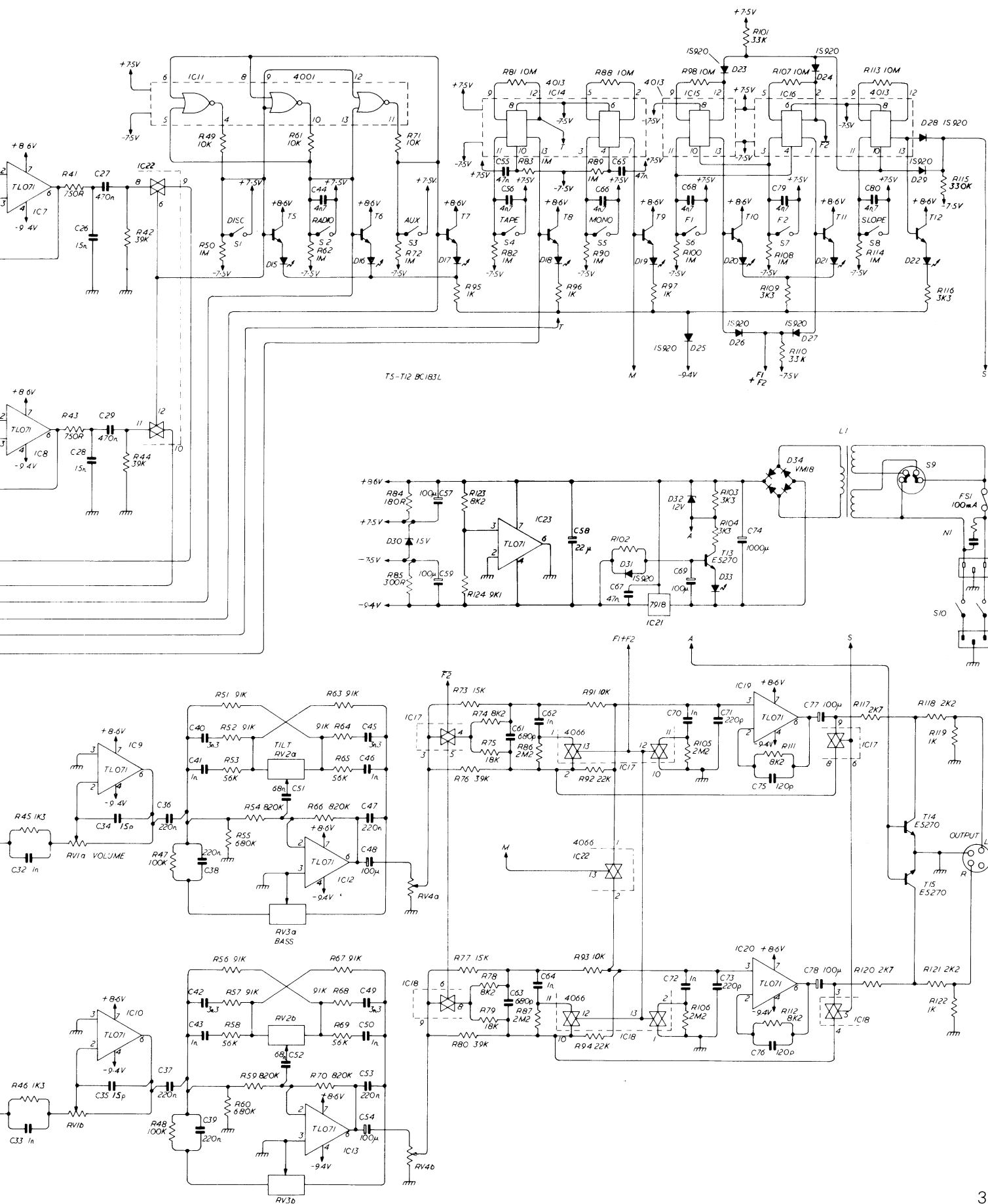
# circuit diagram

## M12746 ISS 1

Up to S/N 6000 PCB M12730 ISS 1 and 2  
for minor variations see 'modifications'

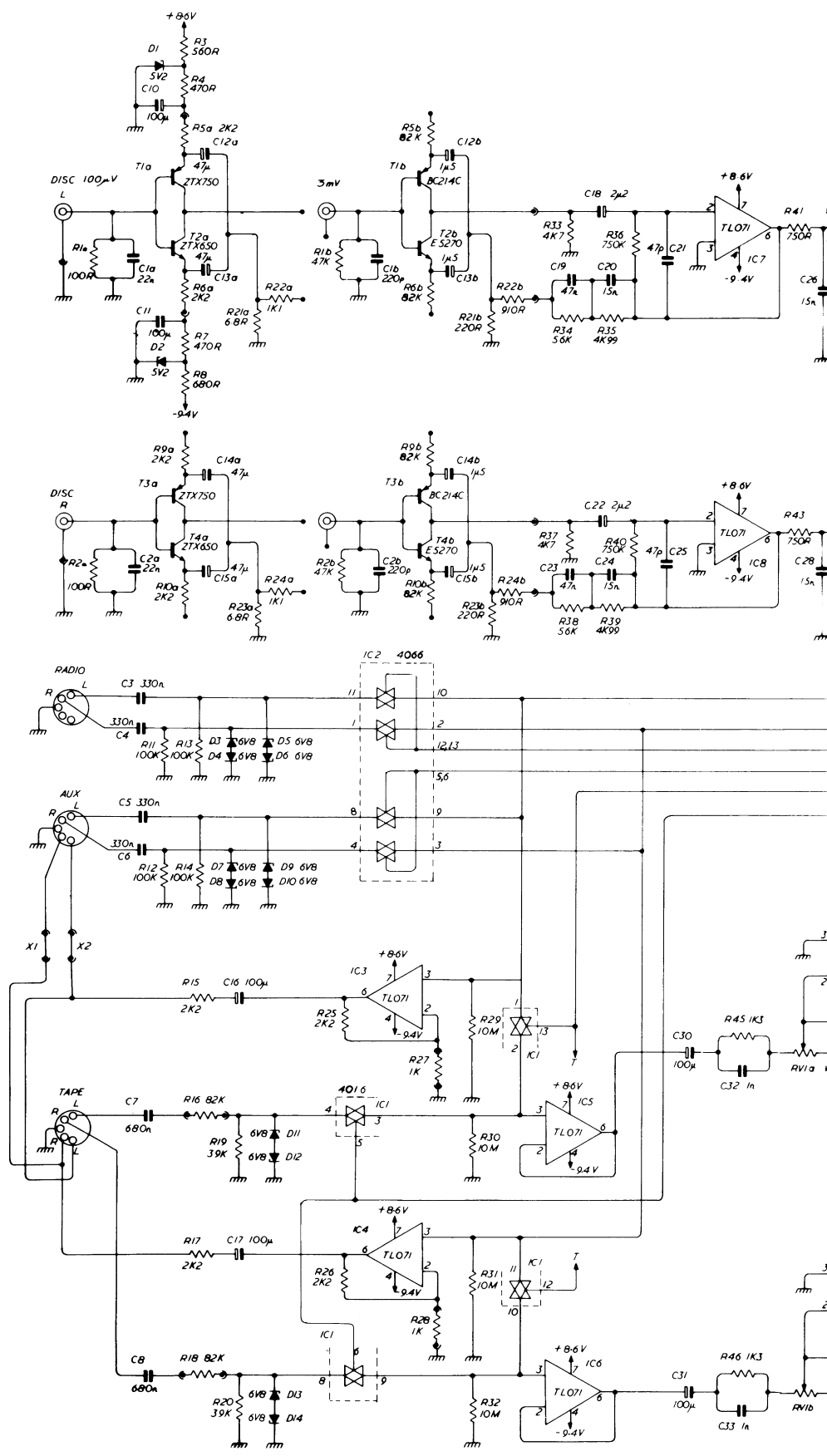


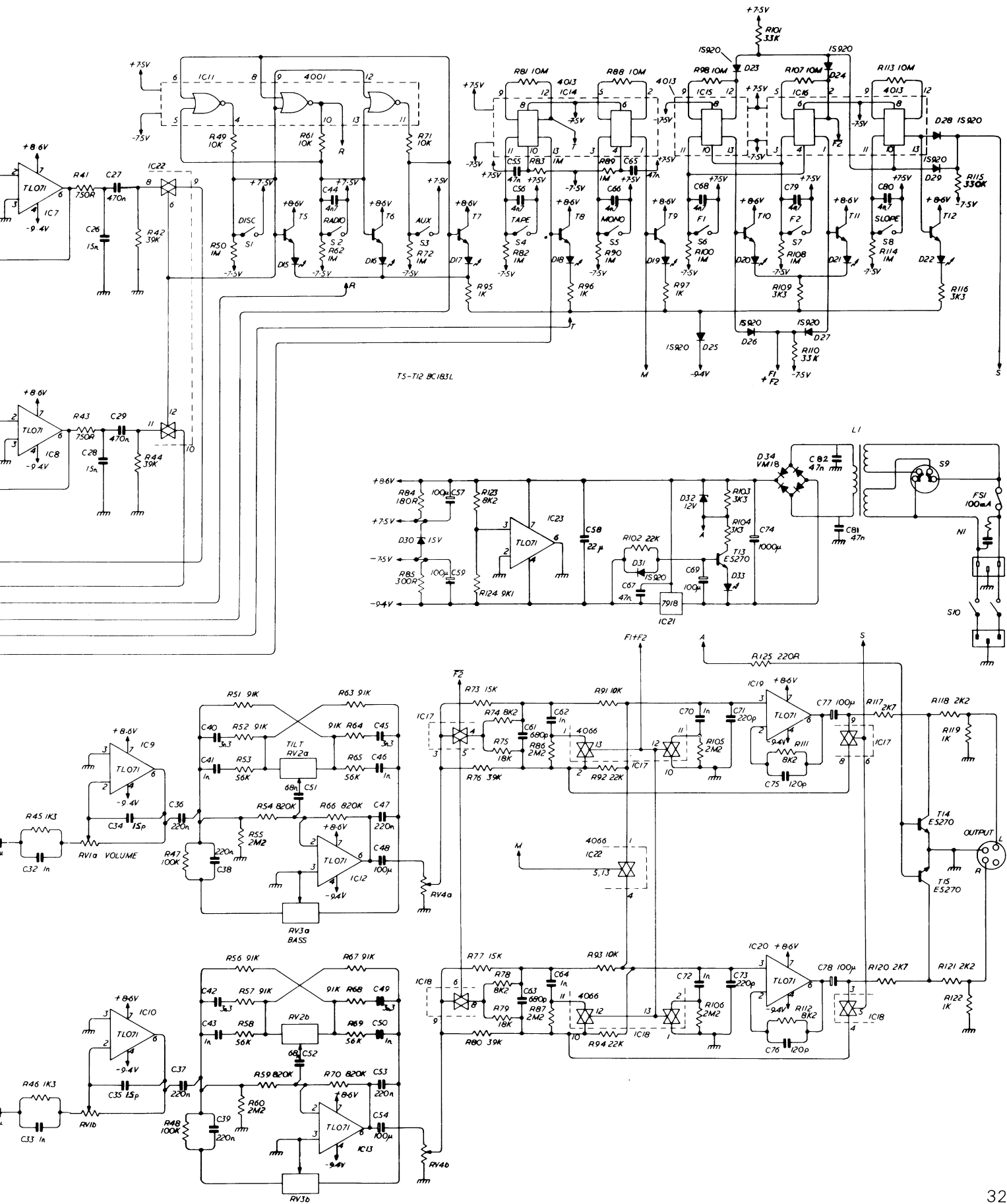
# QUAD 34 service data



# circuit diagram

**M12746 ISS 2**  
S/N 6001 to 8000 PCB M12730 ISS 4  
for minor variations see 'modifications'





# circuit diagram

## M12746 ISS 3

From S/N 8001 PCB M12730 ISS 5  
for minor variations see 'modifications'

